

Pharmaceutical applications of 1,4-benzodiazepines

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5.1 Introduction

Benzodiazepines are one of the most prescribed medicines in the US alone – for example, approximately 65.9 million office-based physician visits, or 27 per 100 adult patients, resulted in a prescription for benzodiazepines each year between 2014 and 2016 (Santo et al., 2020). Since the emergence of the first 1,4-benzodiazepine, chlordiazepoxide (*Librium*) in 1957, a large number of analogs have been synthesized and evaluated in biology and medicine. The increased clinical efficacy and relatively minor side effects

Table 5.1 Overview of FDA-approved short-acting 1,4-benzodiazepine-based drugs.

Generic name	Brand name	FDA-approved indications	Approval date	Manufacturer
Midazolam	Versed	Seizures, sedative	1985	Roche, HLR
Triazolam	Halcion	Insomnia	1982	Pharmacia & Upjohn

over other pharmaceutical agents have rendered 1,4-benzodiazepines the medicine of choice in the treatment of anxiety, insomnia, epilepsy, muscle spasms, alcohol withdrawal, and dementia (Griffin et al., 2013; Atkin et al., 2018; Sachdeva et al., 2015). As discussed in Chapter 4, benzodiazepines decrease neuronal stimulation to give a sedative effect, which is thought to be a result of γ -aminobutyric acid (GABA) receptor binding, enhancing the effect of GABA and increasing cellular chloride influx (Rudolph et al., 1999). While a comprehensive understanding of their pharmacology (including efficacy and side effects) can inform their pharmaceutical applications, 1,4-benzodiazepines and their active metabolites are often grouped on the basis of their duration of action (described by half-life, $t_{1/2}$). In this chapter, the pharmaceutical applications of FDA-approved 1,4-benzodiazepines is discussed, covering short-acting ($t_{1/2} = 1\text{--}8$ h), intermediate-acting ($t_{1/2} = 8\text{--}40$ h) and long-acting ($t_{1/2} = 40\text{--}200$ h) types.



5.2 Short-acting 1,4-benzodiazepines

Short-acting 1,4-benzodiazepines are often used to treat sleep disturbance, taken in low doses three to four times a day. With a half-life of less than 8 h, the risk of overdose is markedly reduced. These medicines are practical for mitigating immediate, one-off cases of stress and anxiety in a PRN (*Pro Re Nata*, or “taken as necessary”) fashion (Starcevic, 2012). The risk of dependence to short-acting 1,4-benzodiazepines is significantly high if taken for more than three weeks, where withdrawal symptoms include sleep disturbance, irritability, and difficulty concentrating and can be eased via careful monitoring and management by health professionals. Table 5.1 gives an overview of two commercial short-acting 1,4-benzodiazepine-based drugs, which will be discussed in the following sections.

5.2.1 Midazolam

Introduction: Midazolam, often sold under the brand name *Versed*, belongs is a short-acting 1,4-benzodiazepine. Initially approved by the FDA in 1985,

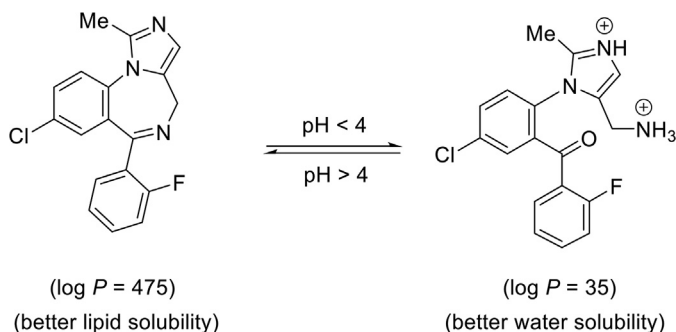


Figure 5.1 Reversible ring-opening of midazolam at different pH conditions.

it can be administered orally, rectally, intranasally, or as an intramuscular (i.m.) or intravenous (i.v.) injection and is used in various clinical settings such as dentistry, cardiac surgery, and endoscopic procedures as a pre-anesthetic and an adjunct to local anesthesia. In late 2018, i.m. administration of this drug was approved by the FDA for the treatment of status epilepticus in adults. In May 2019, the nasal spray of this drug was approved for the acute treatment of distinctive intermittent, and stereotypic seizure episodes in patients whose ages are older than 12 years old. Due to the low risk of dependence and abuse of midazolam, this medication is classified as a schedule IV drug in the United States (Wright et al., 1990).

Midazolam is usually employed for conscious sedation in emergency situations to relieve anxiety and to provide either analgesia or anesthesia during orthopedic or surgical procedures (Nordt & Clark, 1997). Due to its safety profile, rapid onset of clinical effects, and ultra-short length of action (1–3 h), this unique fused heterocycle is also utilized in endoscopy and oral surgeries (Cole et al., 1983; Milgrom et al., 1993). Midazolam is a white to a light yellow crystalline compound that has no solubility in water. The imidazole unit in the midazolam structure is rather basic, which allows the preparation of water-soluble midazolam salt as midazolam hydrochloride ($\text{C}_{18}\text{H}_{13}\text{ClFN}_3 \cdot \text{HCl}$). Under acidic conditions ($\text{pH} < 4$), the diazepine ring of midazolam undergoes a ring-opening reaction that leads to improving its water solubility with the partition coefficient between *n*-octanol and phosphate buffer (log P) of approximately 35 (Fig. 5.1) (Dundee et al., 1984). This water solubility eliminates organic solvents (e.g., propylene glycol) for the medicine packaging and minimizes both pains on injection and the incidence of thrombophlebitis with this medicine.

Pharmacodynamics: Like other benzodiazepines, midazolam cannot directly activate GABA_A receptors, but synergizing the impact of

neurotransmitter GABA (γ -aminobutyric acid) on the GABA_A receptors which results in neural inhibition. This leads to sedation reduction in anxiety, induction of sleep, muscle relaxation anterograde amnesia, and anticonvulsant effects.

Metabolism: Upon entering the bloodstream, midazolam is totally converted to its ring-closed form at the physiological pH and its lipid solubility significantly increases, showing an *n*-octanol/phosphate buffer partition coefficient ($\log P$) of 475. This phenomenon results in the rapid onset of action of midazolam in the body (Gerecke, 1983). Furthermore, the metabolism of midazolam undergoes hepatic CYP450 enzymes and glucuronide conjugation.

Pharmacokinetics and dosage: Regardless of the administration route, midazolam after absorption needs to pass a route to reach the target tissues to release its bioactivity in the patient's body. Accordingly, the drug bioavailability (F) and maximum drug concentration in plasma ($C_{\max}$) are two underlying parameters to evaluate the effectiveness of the drug in terms of pharmacokinetics. In detail, $C_{\max}$ means the highest drug concentration after administration, and bioavailability (F) reveals the fraction of administered drug that successfully reaches the systemic circulation while remaining chemically intact. In contrast to diazepam and chlordiazepoxide, which have painful and partial intramuscular absorption, midazolam can be quickly absorbed through an intramuscular injection. Although midazolam can be easily absorbed from the gastrointestinal (GI) tract after oral administration, its bioavailability is limited to around 40% due to the first-pass metabolism, which is affected by the inconsistency of intestinal blood flow. This bioavailability is significantly enhanced up to 90% when it is taken via an intramuscular route (Table 5.2) (Nordt & Clark, 1997; Dundee et al., 1984; Clausen et al., 1988). In this case, it is important to mention that the midazolam bioavailability of nasal administration mainly depends on its formulation. For example, the intranasal bioavailability of midazolam is found to be 60% when a propylene glycol-based solution of midazolam is taken nasally (Table 5.2), however, 73% bioavailability is determined when administered in an aqueous cyclodextrin solution (Loftsson et al., 2001) (Wermeling et al., 2009). Likewise, the maximum plasma concentration ($C_{\max}$) and time to peak ($T_{\max}$) of midazolam are dependent upon the medicine dosage and the type of administration (Malinovsky et al., 1993; Crevoisier et al., 1983; Hung et al., 1996). For example, $C_{\max}$ values reach 182 ng/L and 48 ng/L within around 12 and 13 min, respectively, after the nasal and rectal administration of 0.2 mg/kg in children (Malinovsky et al., 1993).

Table 5.2 Dosage and pharmacokinetic profile of midazolam.

Administration	Commercial form	Recommended dose (mg/kg)	Onset of action (min)	Bioavailability (%)
Intravenous	Ampoule (1–5 mg/mL)	0.05–0.1	2–3	100
Intramuscular	Ampoule (1–5 mg/mL)	0.05–0.1	5	~ 90
Oral	Tablet or syrup (7.5–15 mg)	0.3–0.75	15	~ 40
Nasal	Spray (5–20 mg/mL)	0.4–1.0	15	~ 60 ^a
Rectal	Ampoule (1–5 mg/mL)	0.2–0.3	10–15	~ 50

^a Bioavailability increases to 73% when administered in an aqueous cyclodextrin solution (17 mg/mL).

Midazolam can be administered through intravenous (i.v.), intramuscular (i.m.), nasal, oral, and rectal routes (Table 5.2), with mostly i.v. and i.m. uses in adults. The recommended midazolam dosage for both i.v. and i.m. administration is approximately 0.05–0.1 mg/kg, depending on the nature of the clinical procedure (Kupietzky & Houpt, 1993). A higher dosage ranging between 0.3–0.75 mg/kg is normally required for the oral route to provide similar clinical effects as the i.v. route. Nasal and rectal midazolam are commonly used for children with the optimal sedative doses of 0.4–1 mg/kg and 0.2–0.3 mg/kg for the former and the latter administration, respectively (Kupietzky & Houpt, 1993). Midazolam is commercially available in sterile, nonpyrogenic ampoules with various volumes (1–10 mL) ready for either intramuscular or intravenous injection. Each vial has midazolam hydrochloride equivalent to 1 mg/mL or 5 mg/mL of midazolam, mixed with sodium chloride (0.8%), edetate disodium (0.01%), and benzyl alcohol (1%). Midazolam is also available in tablet or syrup forms for oral administration. While the dosage ranges in tablets are between 7.5 and 15 mg, the syrup contains 2 mg/mL midazolam in the form of midazolam hydrochloride with a pH of 2.8–3.6, adjusted by hydrochloric acid. This medicine shows a rapid onset of action within 2–3 min after intravenous injection and within 15 min following oral, nasal, and rectal administration (Nordt & Clark, 1997).

Adverse effects: The use of midazolam during the pregnancy is unsafe and it can be recommended only when there is no other alternative for the patient. If this medicine is used during pregnancy, or pregnancy occurs while taking this medicine, the patient should be informed of possible side effects that may threaten the health of the fetus (Kronenfeld et al., 2017). Moreover, infants exposed to high doses of midazolam in the last

weeks of pregnancy may face some serious health issues including fetus heart irregularities, hypotension, poor sucking, hypothermia, and respiratory problems. The baby should also be monitored for signs of acute postpartum withdrawal syndrome. The clinical examinations on humans show that benzodiazepine-related teratogenicity and congenital anomalies can occur as its side effect. However, no information has been provided on the adverse effects of midazolam use in the third trimester of pregnancy on fetus brain development. This drug belongs to group C of AU TGA of pregnancy, which means that it is one of the drugs that can be dangerous for the fetus and cause abnormalities for the baby. Moreover, since this drug can be released in breast milk, it is advised to use it with caution (Nordt & Clark, 1997). For example, some experts recommend waiting at least 4 hours after an i.v. injection for breastfeeding. The American Academy of Pediatrics (AAP) has declared that midazolam with prolonged exposure has an unknown effect on infants but may be worrying. In addition, the use of midazolam for people with obesity may have moderate danger by distributing in the fatty tissues of the patient, prolonging the half-life ($t_{1/2}$), and eventually intensifying the side effects of midazolam.

Interactions: Currently, 464 drugs have been reported that can interact with midazolam. In this case, while 34 drugs have major interactions, 391 and 39 drugs display moderate and minor interactions, respectively. Aluminum hydroxide, baclofen, betamethasone, bupropion, calcium carbonate, codeine, darunavir, efavirenz, fluconazole, ketamine, and lonafarnib are some examples that may interact with the midazolam (Ahonen et al., 1997; Dirig & Yaksh, 1995; Ma et al., 2015). Alcohol can also interact with midazolam and therefore, the simultaneous use of them can enhance the feeling of drowsiness, tiredness, and dizziness. There are no major interactions reported for the simultaneous use of midazolam and fruits, except for grapefruit, which may cause some potential side effects (Cheng et al., 2002).

5.2.2 Triazolam

Introduction: Triazolam (as a generic name) under the brand name of *Halcion* and with IUPAC name of 8-chloro-6-(*o*-chlorophenyl)-1-methyl-4*H*-triazolo-[4,3-*a*]-[1,4]-benzodiazepine is a white crystalline powder broadly administrated for the oral treatment of severe or chronic insomnia. Triazolam is a chemical analog of alprazolam with the difference of having a chlorine atom in the *ortho* position of the 6-phenyl ring, which was approved by the FDA as a short-term hypnotic agent in 1982 (Abraham & Sheppard, 1999; Cairns et al., 1994).

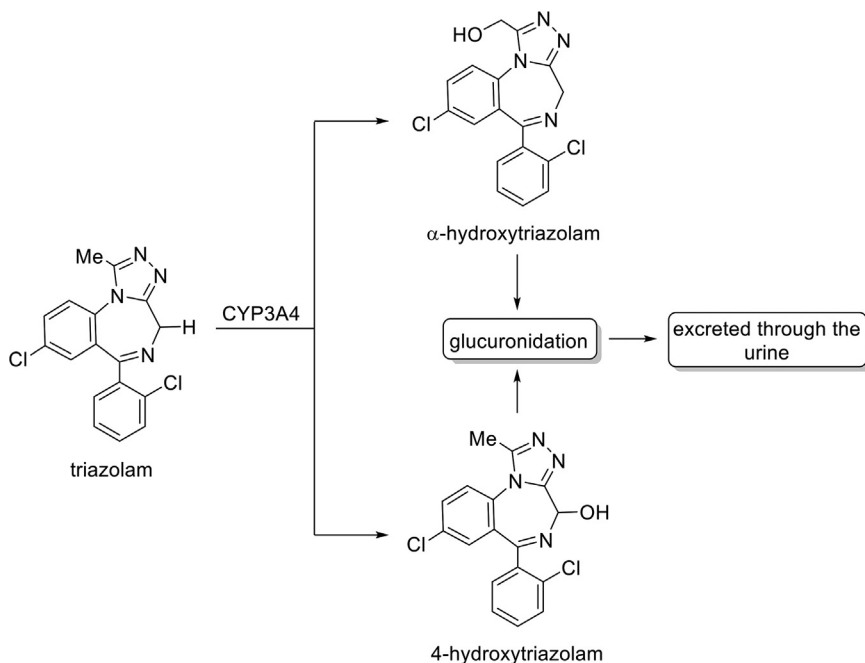


Figure 5.2 Metabolic pathway of triazolam.

Pharmacodynamics: Triazolam is extremely advised by the FDA to be administered at low doses with appropriate care and labeling (Pakes et al., 1981). Similar to most benzodiazepines, triazolam also binds to benzodiazepine receptors, which are classified as BNZ1 receptor being responsible for mediating sleep, and BNZ2 receptor for muscle relaxation, anticonvulsant activity, motor coordination, and memory. Upon the formation of triazolam-receptor complexes, they are coupled to γ -aminobutyric acid A (GABA_A) receptors, which eventually enhances the affinity of GABA neurotransmitters to the receptors. Binding the GABA neurotransmitter to the specific site of GABA opens the chloride channel and results in a hyperpolarized cell membrane that prevents further excitation of the cell (Zhang & Jackson, 1993).

Metabolism: Similar to other benzodiazepines, triazolam metabolizes in the human liver by CYP3A4 enzyme into two main metabolites, *i.e.*, α -hydroxytriazolam and 4-hydroxytriazolam. The drug can be excreted through the urinary system, mainly as glucuronide-conjugated α -hydroxy and 4-hydroxy metabolites, which are not assumed as active metabolites (Fig. 5.2) (Eberts et al., 1981).

Table 5.3 Pharmacokinetic parameters (single 0.88 mg dose) of triazolam.

Plasma albumin binding	89%
Maximum plasma concentration ($C_{\max}$)	8.8 ng/ml
Time to peak drug ($T_{\max}$)	1.3 h
Volume of distribution (V_d)	1.1 L/kg
Elimination half-life ($t_{1/2}$)	2.2–2.7 h

Table 5.4 Administration profile of triazolam.

Insomnia	Adults (< 61 years old):	Geriatrics (> 61 years old):
	0.25–0.5 mg/day	0.125–0.25 mg/day

Pharmacokinetics and dosage: Triazolam with the onset time of 15–30 min, followed by a quick elimination with a half-life ($t_{1/2}$) of 2–5 h, is known as one of the short-acting commercially available benzodiazepines (Roth et al., 1983). Upon the oral administration of 0.88 mg triazolam, the drug quickly reaches out to the central nervous system (CNS) due to its high lipophilicity, exhibiting a maximum plasma concentration ($C_{\max}$) of 8.8 ng/mL after 1.3 h (Table 5.3). Triazolam has relatively a shorter half-life in comparison with chlordiazepoxide, diazepam, and flurazepam, where more than 90% of the drug can be excreted from the circulatory system within 72 h. Triazolam dosage as oral tablets for adults and geriatrics are 0.25–0.5 mg/day and 0.125–0.25 mg/day, respectively, with a maximum level of 1.0 mg/day (Table 5.4). On the basis of a minimum administrated dosage, triazolam is significantly more potent than other benzodiazepines in the treatment of insomnia (Pakes et al., 1981). For example, a study of 20 male patients with insomnia prescribed with triazolam and flurazepam revealed that 1.0 mg of triazolam had a significantly superior impact on inducing sleep compared to 30 mg of flurazepam (Wang & Stockdale, 1973).

Adverse effects: Currently, more than 30 hypnotics are available in the market for the treatment of insomnia. Most of them, such as barbiturates, meprobamate, and diphenhydramine, cause several side effects, e.g., high risk of addiction and overdose, and frequent paradoxical effects in the elders (Dawson-Butterworth, 1970; Lamy & Kitler, 1971; Piccione et al., 1980), while triazolam seems to be a promising alternative by exhibiting better tolerance, less toxicity, and lower risk of overdose (Murphy et al., 1982). Additionally, the major side effects of hypnotic benzodiazepines are associated with a daytime hangover due to the remaining drug in the circulatory system during the daytime. However, the short-acting triazolam could be the best choice for the treatment of insomnia in geriatrics because of its quick elimination from the body. On the downside, triazolam has been

associated with numerous reports of retrograde amnesia (Bixler et al., 1991). Triazolam may have some further minor side effects such as drowsiness, headache, dizziness, nervousness, and laziness. Importantly, its interaction with some other drugs such as nefazodone, antibiotics, antifungals, anti-HIV, anti-hepatitis C, and codeine or hydrocodone drugs can cause serious issues such as the risk of severe breathing disorders or coma. Thus, precautions for the safe coadministration of triazolam are highly advised. Another precaution for triazolam use is associated with the pregnancy and lactating period of females. Triazolam is classified as a category X drug by the US FDA for pregnant females since it can trigger fetus abnormalities and serious birth defects. Infants exposed to triazolam via breast milk should be controlled for respiratory depression, sedation, withdrawal symptoms, and feeding problems (van Strien et al., 2013). It is therefore highly offered that a lactating woman should consider pumping and discarding breast milk 28 h after triazolam administration to reduce the risk of drug exposure to her infant.

Interactions: Triazolam, same as other benzodiazepines, is a highly interactive drug with many other medications, in which coadministering with other drugs significantly diminishes the impact of triazolam, generating some mild to serious side effects in patients. Examples of deactivating agents include boceprevir, mifepristone, telaprevir, antidepressants (e.g., nefazodone and fluoxetine), azole antifungals (e.g., itraconazole and ketoconazole), HIV protease inhibitors (e.g., lopinavir and ritonavir), rifamycins (e.g., rifampin, rifabutin), macrolide antibiotics (e.g., erythromycin and clarithromycin) (Yuan et al., 1999; Phillips et al., 1986).



5.3 Intermediate-acting 1,4-benzodiazepines

Intermediate-acting 1,4-benzodiazepines with an elimination half-life of 8–40 h are usually administered for the treatment of anxiety and insomnia (Leung, 2011). These medications may have some next-day residual effects such as drowsiness, dysarthria, and fatigue when taken for hypnotic purposes, in which the total sleep time is increased via reducing the number of awakenings. Compared with the short-acting benzodiazepines such as triazolam for treating insomnia, the intermediate-acting analogs are more effective for those patients suffering from awaking in the middle of the night and with difficulties of falling asleep again. To be more precise, short-acting benzodiazepines are generally prescribed for initiating sleep in insomniacs, while intermediate-acting agents are administered for difficulties with sleep maintenance. 1,4-Benzodiazepines with the intermediate

Table 5.5 Overview of FDA-approved intermediate-acting 1,4-benzodiazepine-based drugs.

Generic name	Brand name	FDA-approved indications	Approval date	Manufacturer
Estazolam	ProSom	Insomnia	1990	Abbott
Lorazepam	Ativan	Anxiety, seizures sedation	1977, 1980	Wyeth, Valeant
Temazepam	Restoril	Insomnia	1981	Tyco Healthcare
Alprazolam	Xanax, Niravam	Anxiety, depression	1981, 2005	Pfizer, Schwarz Pharma

therapeutic acting should preferably be taken about once or twice a day, which is frequently used for long-term treatment. Similar to other 1,4-benzodiazepines, these drugs have continuous popularity in the management of anxiety disorders due to their consistent effectiveness, relative safety in overdose, rapid onset of action, and the possibility of administration on a PRN (as needed) basis (Starcevic, 2012). On the contrary, abrupt withdrawal of intermediate-acting 1,4-benzodiazepines may cause severe withdrawal symptoms, which appeared in both physical and psychological symptoms such as seizures, weakness, muscle tension, panic disorders, agitation, hallucinations, and delirium (Soyka, 2017). Therefore, the withdrawal procedure must be considered to avoid the withdrawal symptoms with a slow and gradual reduction of the drug dosage. Table 5.5 outlines the most common FDA-approved intermediate-acting 1,4-benzodiazepine-based drugs, which will be discussed in this section.

5.3.1 Estazolam

Introduction: Estazolam (as a generic name) with the US brand name of *ProSom* and IUPAC name of 8-chloro-6-phenyl-4*H*-[1,2,4]triazolo[4,3-*a*][1,4]benzodiazepine is a triazole-fused benzodiazepine approved by the FDA for the treatment of insomnia in 1990. In general, estazolam has anxiolytic, anticonvulsant, hypnotic, sedative, and skeletal muscle relaxant properties prescribed for the remedy of insomnia by decreasing the number of awakenings and improving sleep maintenance during a night (Post et al., 1991; Roehrs et al., 1983; Lamphere et al., 1986).

Pharmacodynamics: Estazolam is a 1,4-benzodiazepine-based drug derived from triazolobenzodiazepines that binds to the benzodiazepine receptors in GABA postsynaptic neurons in several parts of the central nervous system (Braestrup & Squires, 1978). This phenomenon leads to

Table 5.6 Pharmacokinetic parameters of estazolam.

Plasma albumin binding	93%
Time to peak drug ($T_{\max}$)	0.5–6 h (average = 2 h)
Elimination half-life ($t_{1/2}$)	10–24 h
Excretion	Urine (> 70% as inactive metabolites), Feces (4%)
Bioavailability	93%
Duration of action	Variable

Table 5.7 Administration profile of estazolam.

Insomnia	Adults (< 61 years old): 1–2 mg orally at bedtime	Geriatrics (> 61 years old): 1 mg orally at bedtime
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increasing the permeability of the neuronal membrane to chloride ions and inducing relaxation, sleep regulation, muscle relaxant, and anticonvulsant. In the case of insomnia treatment, taking estazolam at high doses causes a significant decrease in histamine release in the brain via having an effect on the benzodiazepine–GABA receptor complex, which promotes NREM (non-rapid eye movement) sleep by shutting down the cortex and thalamus (Oishi et al., 1986).

Metabolism: Hepatic metabolism of estazolam extensively occurs in the liver to its 11 metabolites, mainly as 4-hydroxyestazolam, 1-oxo-estazolam, 4'-hydroxyestazolam, which are then excreted in the urine. The main metabolite of estazolam, 4-hydroxyestazolam, is found to be a slightly active metabolite, where its high accumulation in the body may cause drug overdose (El-Haj et al., 2018).

Pharmacokinetics and dosage: Estazolam is commercially available as oral tablets (1 and 2 mg), displaying a rapid onset time of 15–30 min with $t_{1/2}$ = 10–24 h. By using estazolam, total sleep time in patients can increase up to an average of 63 minutes on the first night of therapy, with the peak plasma level ranging 1–6 h. Owing to the relatively long half-life of estazolam, it can be also prescribed for insomniacs who are concomitant with general anxiety disorders. For example, Post and co-workers analyzed 99 randomly selected patients administered either a 2.0 mg dose of estazolam or equivalent placebo for 7 nights. The results indicated that the hypnotic impact of estazolam was statistically remarkable compared to placebo in treating the sleep disorder cases ($p < 0.01$) (Post et al., 1991). Pharmacokinetics parameters and recommended dosages of estazolam are shown in Tables 5.6 and 5.7, respectively (Aoshima et al., 2003; Gustavson & Carrigan, 1990).

Adverse effects: Somnolence, hypokinesia, sleepiness, dizziness, and abnormal balance are the main side effects of estazolam (Pierce et al., 1990). Although the adverse effects of benzodiazepines are generally higher for geriatrics, estazolam is safer and more effective for this group of patients (> 65 years old) in the treatment of insomnia (Vogel & Morris, 1992). While the long- and short-acting benzodiazepines prescribed for insomnia frequently disclose anterograde amnesia (i.e., a loss of memory for recent events) and daytime sedative effects, respectively, neither of them has been observed for estazolam. Nevertheless, estazolam should be administered with carefulness to avoid any possible negative effects. The major concerns of taking estazolam are the risks of drug dependency and addiction, in which abrupt discontinuation of the medication may cause subsequent withdrawal syndrome such as rebound insomnia (Carico et al., 2018). Therefore, it is highly recommended to take estazolam for a short-term treatment with the lowest effective dosage to prevent physical or behavioral dependence. The misuse of estazolam may proceed to exanimation, mydriasis, stunning, or even death. It has been observed in some cases that taking estazolam causes allergies, which is not undeniable. Patients should not use this medicine if they have sleep apnea. Sleep apnea is a condition that the muscles and tissues around the throat relax and block the passage of air, causing the person to stop breathing for 10 seconds or more during sleep. Asthma and chronic obstructive pulmonary disorders are some of the diseases that should not be treated with estazolam. Furthermore, people with liver and kidney problems should not take estazolam for the treatment of insomnia because they can hardly excrete estazolam from the body, leading to an overdose. Estazolam should not also be administrated in the cases of pregnancy and during breastfeeding, where it causes birth defects and inevitable symptoms. In a study carried out by Iqbal and co-workers, it was shown that a fetus fed by a mother using estazolam exhibited several withdrawal symptoms (Iqbal et al., 2002).

Interactions: Estazolam has a potential drug-drug interaction with 28 medications, in which the simultaneous consumption of estazolam with them may deal with several problems such as drowsiness and slow breathing. For example, ritonavir (used for the treatment of HIV/AIDS), itraconazole, and ketoconazole have serious interactions with estazolam, thus, their concurrent administrations should be avoided. The simultaneous use of alcohol may have no serious interactions but can intensify the sedative and hypnotic effects of estazolam during the treatment.

5.3.2 Lorazepam

Introduction: Lorazepam (as a generic name) under the brand name of *Ativan* and IUPAC name of 7-chloro-5-(2-chlorophenyl)-3-hydroxy-1,3-dihydro-1,4-benzodiazepin-2-one, is a white powder that is relatively soluble in water, known as a drug for treating insomnia, anxiety, and epilepsy. It is categorized as a Class II drug that is available both as oral tablets and in injection. Lorazepam was initially developed and marketed by Wyeth Pharmaceuticals under the brand name *Ativan* and *Temesta* in the US in 1977, followed by the FDA label approval in 1985 ([The US Food and Drug, 2019](#)).

Pharmacodynamics: Lorazepam is an anxiolytic, sedative, hypnotic, anti-amnesia, anticonvulsant, and muscle relaxant medicine ([The US Food and Drug, 2019](#)). Similar to other benzodiazepines, lorazepam relaxes the body via binding to the benzodiazepine receptors in GABA postsynaptic neurons in the central nervous system, particularly in the subcortical and limbic areas. This medicine also stimulates GABA receptors in the ascending reticular activating system (ARAS), modulating wakefulness and sleep-wake cycles ([Gupta et al., 1990](#); [Siepmann et al., 2007](#); [The US Food and Drug, 2019](#)).

Metabolism: Lorazepam is extensively bound to plasma proteins and metabolized in the liver, in which about 95% of orally administered lorazepam is excreted in the urine and feces within 5 days. Three main metabolites are generally formed 4 h after administration of lorazepam including the lorazepam-glucuronide, metabolites I, and II ([Fig. 5.3](#)) ([Elliott, 1976](#)). The concentration and types of metabolites highly depend on the animal species.

Pharmacokinetics and dosage: Lorazepam is administered orally, intravenously (i.v.), and intramuscularly (i.m.). The onset time of intravenous, intramuscular, and oral administrations of lorazepam are 1–3 min, 15–30 min, and 20–30 min, respectively ([Greenblatt et al., 1982](#); [Greenblatt et al., 1982](#)). Lorazepam can be readily absorbed with an absolute bioavailability of 90% and the mean half-life of unconjugated lorazepam in human plasma is approximately 12 h ([Table 5.8](#)). Peak concentrations in plasma happen about 2 h after oral administration. The peak plasma level of lorazepam for the 2 mg dose is approximately 20 ng/mL. Since lorazepam has high lipophilicity, its rectal administration is not practical ([Greenblatt et al., 1979](#)). The usual bedtime dose of lorazepam is 2–6 mg/day given in divided doses; however, the daily dosage could vary from 1 to 10 mg daily

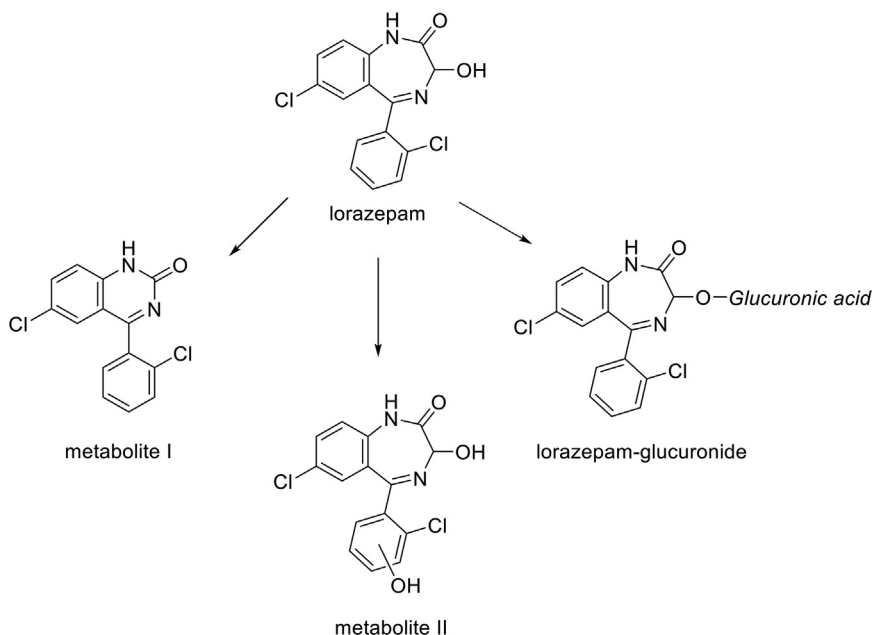


Figure 5.3 Three main metabolites of lorazepam.

Table 5.8 Pharmacokinetic parameters of lorazepam.

Plasma albumin binding	91%
Bioavailability	90%
Time to peak drug ($T_{\max}$)	≤ 3 h (i.m.), ~ 2 h (oral)
Volume of distribution (V_d)	(a) Neonates (i.v.): 0.14–1.3 L/kg (average: 0.76 L/kg); (b) Pediatric (i.v.): for 5 months to < 3 years: 0.67–3.4 L/kg (average: 1.62 L/kg), for 3 to < 13 years: 0.49–3 L/kg (average: 1.5 L/kg), for 13 to < 18 years: 1–1.54 L/kg (average: 1.27 L/kg); (c) Adults: 1.3 L/kg
Elimination half-life ($t_{1/2}$)	(a) Neonates (i.v.): 18–73 h (average: 40 h); (b) Pediatric (i.v.): for 5 months to < 3 years: 6–28 h (average: 16 h), for 3 to < 13 years: 7.5–41 h (average: 17 h), for 13 to < 18 years: 8.2–42 h (average: 18 h); (c) Adults: ~ 12 h (oral), ~ 14 h (i.v.), ~ 13 –18 h (i.m.)
Excretion	Urine ($\sim 88\%$, predominantly as inactive metabolites), Feces ($\sim 7\%$)
Onset of action	Anticonvulsant (i.v.): 10 min; Hypnosis (i.m.): 20–30 min; Sedation (i.v.): 15–20 min; Sedation (oral): 20–30 min
Duration of action	Adults (i.m. and i.v.): 6–8 h

Table 5.9 Administration profile of lorazepam.

Age	Insomnia	Anxiety	Light anesthesia	Status epilepticus
Adults	2 mg/day (oral)	2–3 mg/day (oral)	2 mg (i.m.)	2 mg (i.m.)
Geriatrics	2–4 mg/day (oral)	1–2 mg/day (oral)	–	–
Pediatrics (≥12 years old)	2–4 mg/day (oral)	–	–	–

(Riss et al., 2008). The size and duration of the drug effects in the body are highly dose dependent. Commercially available oral tablets contain 0.5 mg, 1 mg, or 2 mg of lorazepam. Those patients suffering from anxiety should take the initial prescribed dose of 1 mg two to three times daily. Due to transient situational stress or anxiety, insomniacs require a single dose of 2–4 mg at bedtime. For geriatrics, an initial dosage of 1–2 mg/day in divided doses is recommended (Greenblatt et al., 1979). It has been recommended that patients with the experience of alcohol dependence should be carefully administered with the lowest dosage of lorazepam because of the possibility of several harmful consequences such as hostility, seizure, and confusion (Ramanujam et al., 2015). The pharmacokinetic parameters and recommended dosages of lorazepam are shown in Tables 5.8 and 5.9, respectively.

Adverse effects: Lorazepam is classified as a Class IV drug, according to the Drug Enforcement Administration (DEA), which means that administration of lorazepam has a risk of dependence and abuse. Side effects of lorazepam occur on the CNS (i.e., headache, dizziness, fatigue, weakness, drowsiness, nightmare, ataxia, tremor, and depression), GI (i.e., nausea, vomiting, dry mouth, difficulty in swallowing, constipation, and anorexia), and other symptoms such as rash and skin allergies, bradycardia, vascular collapse, urinary retention, binoculars, blurred vision, nystagmus, and respiratory depression. More importantly, abusing or taking high dosages of lorazepam may increase the possibility of death in humans (Kripke, 2016). According to a study on rats, repeated injection of lorazepam may cause drug resistance due to delaying sodium channel regeneration in the rats' spinal cells (McLean & Macdonald, 1988). Furthermore, it has been shown that a long-term administration of lorazepam can arise the incident of memory loss (Bishop & Curran, 1998). Formulations of lorazepam with 2 and 4 mg/mL concentrations have a significant amount of 830 mg/mL propylene glycol.

Accordingly, propylene glycol poisoning often happens in the case of high-dosage or long-term prescriptions of lorazepam (Riker & Fraser, 2005). The US FDA has classified lorazepam as a group C drug for pregnant women, having a high potential of harmful effects on the fetus. For example, the use of lorazepam during pregnancy or childbirth will cause some babies to be urged for respiratory treatment at birth. It should be noted that glucuronidation of this drug may competitively inhibit bilirubin binding that leads to hyperbilirubinemia in infants (Di Michele et al., 1996). According to the latest announcement of the AAP, this drug may result in alarming risks in babies who are breastfed by lorazepam-contaminated milk for a long period of time (Kronenfeld et al., 2017).

Interactions: Lorazepam has been shown to have interactions with aspirin, vitamin C, vitamin B12, and omega 3. This medication can also interact with alcohol to intensify its side effects, appearing in the forms of dizziness, drowsiness, and difficulty in concentrating (Aranko et al., 1985). Obesity can also extend the half-life of lorazepam due to the high lipophilicity of lorazepam, resulting in the high distribution in the body fat and blood plasma.

5.3.3 Temazepam

Introduction: Temazepam (as a generic name) under the commercial name of *Restoril* with the IUPAC name of 7-chloro-1,3-dihydro-3-hydroxy-1-methyl-5-phenyl-2H-1,4-benzodiazepin-2-one is a white crystalline compound displaying slight solubility in water and alcohol. This 1,4-benzodiazepine-based medication was first developed in 1964, and soon after it became one of the most widely marketed drugs for the treatment of insomnia (Mitler, 1981; The US Food and Drug Administration, 2016b).

Pharmacodynamics: Temazepam operates by manipulating GABA within the brain, generating anti-anxiety, sedative, anticonvulsant, and muscle relaxant effects. Abnormal activity of hypothalamic-pituitary-adrenocortical (HPA) system in the brain is demonstrated to be eminently involved in the pathogenesis of major stress and depression. In a study conducted on rats, it was shown that temazepam could serve in a sedative manner to normalize the HPA axis activity directly via modulating GABA_A receptors and indirectly by enhancing the intra-hypothalamic concentrations of vasopressin (AVP) (Welt et al., 2006). Like other benzodiazepines, binding temazepam to the GABA_A receptors site increases the neurotransmission effects of this chemical mediator, opening the chloride channel on the receptors and enhances neuron polarization. Temazepam appears to potentiate the

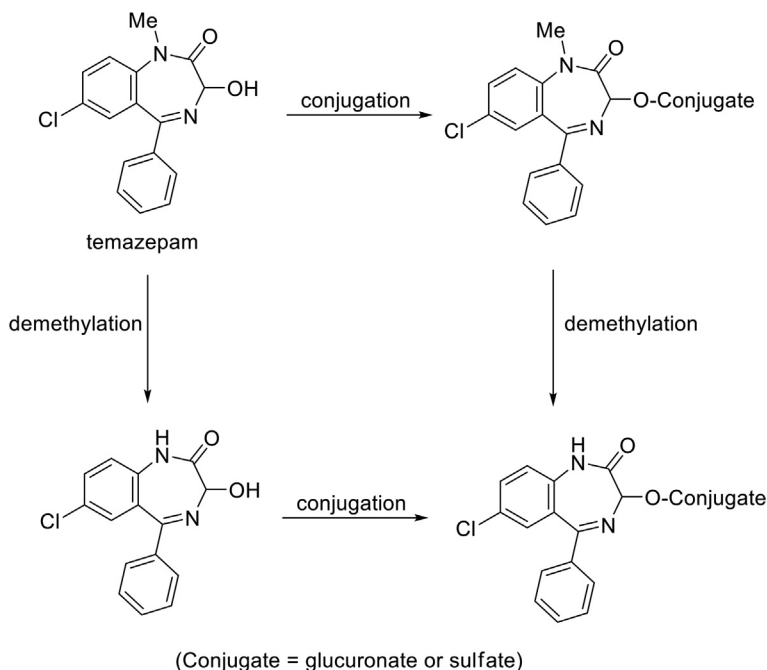


Figure 5.4 Metabolic pathway of temazepam.

inhibitory effect of GABA, which thereby reduces neuronal stimulation in many areas of the brain (Atack, 2003).

Metabolism: Temazepam with 96% plasma protein binding is extensively metabolized in the liver converting to its *O*-conjugates at high concentrations, and only small amounts of the *N*-desmethyl metabolites are found (Fig. 5.4). These four species comprise about 90% of the metabolized temazepam in the blood, which are then completely excreted in the urine and feces (Schwarz, 1979). Note that there is a considerable difference between the metabolic behavior of temazepam in the blood and the urine. For example, the quantity of temazepam in the blood is approximately 36%, while it is reduced to around 2% in the urine. Furthermore, *O*-conjugate and *N*-desmethyl metabolites occur 47% in the blood compared with 96% in the urine. These data strongly suggested that this medication is either conjugated or demethylated before excretion (Schwarz, 1979).

Pharmacokinetics and dosage: Temazepam is pharmaceutically available in either oral capsules or tablets with 7.5–30 mg of the drug. Clinical studies demonstrated that the pharmacokinetics of temazepam depend on the formulations (Salonen et al., 1986; Fuccella, 1979;

Table 5.10 Pharmacokinetic parameters of temazepam after single oral administration of 20 mg.

Oral administration type	C _{max} (ng/mL)	T _{max} (h)	AUG (µg/mL × h)	t _{1/2} (h)
Soft gelatin capsule	0.892	0.83–1.1	3.86–4.23	7–11
Hard gelatin capsule	0.668	1.44–2.8	3.84–4.27	7–11
Tablet	0.284	3.1–4.0	3.95–5.55	7–11

Fuccella et al., 1977). Following ingestion of 20 mg gelatin capsules or tablets by six healthy volunteers, pharmacokinetics parameters of the medication have been compared (Table 5.10). Temazepam is much more rapidly absorbed from the soft gelatin capsule ($T_{\max} = 0.83\text{--}1.1$ h) than from the tablet ($T_{\max} = 3.1\text{--}4.0$ h), due to faster gastrointestinal absorption. There is no significant difference observed in the total AUC and elimination half-life of the three temazepam formulations, indicating that only the rate of absorption and distribution of the drug is dependent on the prescribed dosage form. The recommended usual dosage of temazepam for adult insomniacs is 10 to 20 mg half an hour before bedtime. However, 7.5 mg is suggested as the initial dose for individuals aged 65 or above because of the high risk of oversedation, dizziness, and confusion in elderly patients (The US Food and Drug Administration, 2016b). In a sleep assessment of eight healthy male volunteers, the effects of 10-day administration of 30 mg dosage of temazepam were compared with placebo (Ferrillo et al., 1984). Temazepam was found to significantly reduce the number of nocturnal awakenings, phase shifts, as well as a major increase of duration of stage II and sleep efficiency.

Adverse effects: Laziness, motor disorders, low driving ability, faintness, nystagmus, and the dangers of falling in the elderly are among the most common side effects of taking temazepam (The US Food and Drug Administration, 2016b). The tolerance of these effects may vary in people of different species, ages, and gender. It has been evident that temazepam gradually loses its effectiveness after around 14 days of consumption (Kales et al., 1986a). According to the FDA drug categories, this drug falls into category X, which means that temazepam may cause abnormalities to the unborn baby during pregnancy. Therefore, patients should be advised to stop taking the medication before becoming pregnant. The excretion of temazepam into breast milk during its daily consumption as a sleep-inducing drug in the postpartum period was investigated by Lebedevs and co-workers (Lebedevs et al., 1992). Ten infants (< 15 days old) were breastfed by mothers who had administrated 10 or 20 mg of temazepam (0.16 to 0.32 mg/kg)

at bedtime once daily for at least 2 days. None of the infants had any observable adverse reactions. Temazepam can induce physical dependency and addiction same as other benzodiazepines. Removal of temazepam following daily usage also contributes to the withdrawal syndrome. A slow and careful reduction of the drug dosage is recommended to sustain severe withdrawal symptoms (MacKinnon & Parker, 1982). Rare reports of the relationship between sudden removal of temazepam and psychotic states have been recorded (Terao & Tani, 1988).

Interactions: A variety of antidepressants, anti-anxiety medicines, and opioid analgesics intensify the sedative effects of this medicine. The addition of CNS depressants such as barbiturates, opiates, alcohol, non-selective MAO inhibitors, phenothiazines, and different antipsychotics, skeletal muscle relaxants, and antihistamines can disrupt the function of temazepam. However, the pharmacokinetic interactions of this medication with other co-administered drugs seem to be minimal. Since CYP3A enzymes play a key role in the metabolism of 1,4-benzodiazepines, possible interactions with those drugs having inhibitory effects on the enzymes should be considered. In this case, no noticeable interaction of temazepam with the CYP3A system has been reported (The US Food and Drug Administration, 2016b).

5.3.4 Alprazolam

Introduction: Alprazolam (as a generic name) under the brand names of *Xanax* and *Niravam* with the IUPAC name of 8-chloro-1-methyl-6-phenyl-4*H*-triazolo[4,3-*a*][1,4]benzodiazepine is an intermediate-acting triazolo-fused benzodiazepine, which has been approved for the treatment of anxiety, panic disorders, and depression (Ait-Daoud et al., 2018). Alprazolam was approved by the US FDA in 1981 to be prescribed to treat patients suffering from anxiety or panic disorder. This drug is available as a white crystalline powder that is soluble in methanol and ethanol with no significant solubility in water (The US Food and Drug Administration, 2016d), exhibiting low lipophilicity ($\log P \approx 2.5$, where P is octanol-water partition coefficient). According to clinical data of 84 research studies (Jonas & Cohon, 1993), alprazolam has been shown to be as effective as or even superior to other benzodiazepines and anti-depressants such as diazepam, lorazepam, and bromazepam, clomipramine, dothiepin, amitriptyline, desipramine, doxepin, and imipramine, for the treatment of anxiety disorders.

Pharmacodynamics: The fused triazole ring in the alprazolam structure is believed to play a key role in its adsorption and metabolism. It has been well-understood that therapeutic agents with a fused triazole or

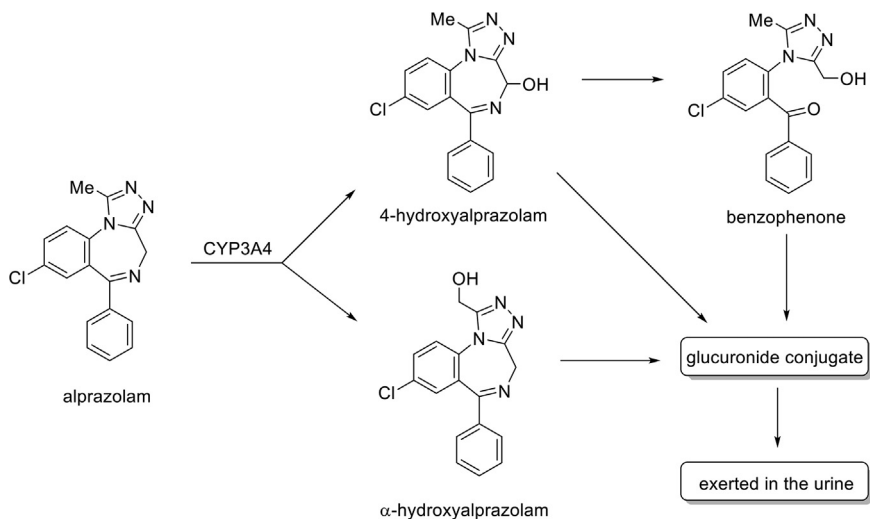


Figure 5.5 Metabolic pathway of alprazolam.

imidazole rings, such as alprazolam and midazolam, are quickly absorbed and metabolized by α -hydroxylation of the methyl group on the triazole or imidazole ring (Ait-Daoud et al., 2018). The presence of the fused triazole ring also provides a strong binding affinity to the GABA_A receptors, enhancing its effectiveness. Alprazolam is also known to be significantly superior to placebo and as effective as tricyclic antidepressants (TCAs) such as clomipramine and imipramine, but with better tolerance and low withdrawal effects (Verster & Volkerts, 2004). Soon enough after alprazolam's medical use approval, it became one of the most widely prescribed psychotropic medications in the US, reporting over 48 million prescriptions in 2013 (Ait-Daoud et al., 2018).

Metabolism: Following oral administration, more than 90% of alprazolam is quickly absorbed from the gastrointestinal tract and binds to plasma albumin proteins, which are mainly metabolized in the liver by the CYP3A4 enzyme. After ingestion of alprazolam, it is converted to two major metabolites, 4-hydroxyalprazolam and α -hydroxyalprazolam, followed by further metabolism of 4-hydroxyalprazolam to its corresponding benzophenone through dissociation of the diazepine ring. Eventually, all metabolites are conjugated with glucuronide and excreted in the urine (Fig. 5.5) (Yasui et al., 1996; von Moltke et al., 1993).

Pharmacokinetics and dosage: Alprazolam's onset time is less than an hour with the elimination half-life ($t_{1/2}$) of 9 to 16 h, depending upon the rate of release of the drug. The duration of action for this drug is approximately

Table 5.11 Pharmacokinetic parameters of alprazolam (single 1.0 mg dose).

Plasma albumin binding	> 90%
Maximum plasma concentration ($C_{\max}$)	12–22 ng/L
Time to peak drug ($T_{\max}$)	1–2 h
Volume of distribution (V_d)	0.8–1.3 L/kg
Elimination half-life ($t_{1/2}$)	9–16 h
Clearance	0.7–1.5 mL/min/kg

Table 5.12 Administration profile of alprazolam.

Treatment	Extended-release tablets		Disintegrating tablets or solution	
	Adults	Geriatrics	Adults	Geriatrics
Anxiety disorders ^a	0.25–0.5 mg (3 times daily)	0.25 mg (3 times daily)	0.25–0.5 mg (3 times daily)	0.25 mg (3 times daily)
Panic disorders ^b	0.5–1 mg (once daily)	0.5 mg (once daily)	0.5 mg (3 times daily)	0.25 mg (3 times daily)

^a Maximum level = 4 mg/day.

^b Maximum level = 10 mg/day.

6 h, where the maximum plasma concentration ($C_{\max}$) reaches 12–22 ng/L within 1–2 h. Pharmacokinetic parameters of alprazolam are highly dose-dependent (Abernethy et al., 1984; Fleishaker & Hulst, 1994), as such, its plasma pharmacokinetic profile after administration of a single 1 mg dose is summarized in Table 5.11. Alprazolam is commercially available either as disintegrating tablets with the dosage ranges from 0.25 mg to 2 mg or an oral solution containing 1 mg/mL, administered in divided doses (The US Food and Drug Administration, 2016d). It is also available in extended-release tablets, enabling the drug to be released in the body gradually after being taken and thereby reducing the frequency of administration. Depending on the type of disorder, age, and the form of medication, various drug dosages are prescribed to the individuals (Table 5.12) (The US Food and Drug Administration, 2016d). Anxiety treatment is often initiated at a low dose of 0.25 mg three times daily for both groups of patients, which may be increased to no more than 4 mg per day based on the drug response (Chouinard et al., 1982). The recommended starting dosages for adults and geriatrics with panic disorders are 0.5 mg/day and 0.25 mg/day, respectively, with a maximum level of 10 mg/day (The US Food and Drug Administration, 2016d; Fawcett & Kravitz, 1982).

Adverse effect: Despite many medical advantages of alprazolam, most addiction specialists have considered alprazolam to have higher misuse liability than other benzodiazepines. In this case, the US National Emergency Department (NED) announced that alprazolam was the second most routine medication and the most common benzodiazepine involved in drug misuse cases (Ait-Daoud et al., 2018). The main reasons for the alprazolam misuse potential may come from its rapid adsorption and metabolism, short half-life ($t_{1/2}$), and high euphoric effects (Verster & Volkerts, 2004), where drug addicts tend to consume the drug for recreational purposes. In terms of medical use, abrupt discontinuation of alprazolam after long-term administration can cause many withdrawal effects such as dizziness, drowsiness, dysarthria, headache, fatigue, memory impairment, and depression. Thus, the drug's daily usage is gradually reduced to zero over a period of several weeks or months to avoid severe quitting consequences (Ait-Daoud et al., 2018). The US FDA has considered alprazolam as a pregnancy category D drug that indicates initial evidence of human fetus risk (The US Food and Drug Administration, 2016d). Approximately one-third of pregnant females are predicted to receive anxiolytics to manage their psychiatric symptoms, which harms the pregnancy (The US Food and Drug Administration, 2016d). Alprazolam and its active metabolite (4-hydroxyalprazolam) can easily pass through the blood-milk barrier into breast milk, displaying similar pharmacokinetics to the drug in the blood plasma of the pregnant mother (Oo et al., 1995). In a MEDLINE search of retrospective studies (1966–2000) investigating pregnancy consequences of females taking alprazolam through the first trimester of pregnancy (Iqbal et al., 2002), those exposed to high doses of alprazolam, e.g., more than 4 mg/day for anxiety and 10 mg/day for panic, exhibited severe growth aberrations and mental disorders in their newborn infants including inguinal hernia, cleft lip, cryptorchidism, hypospadias, tracheoesophageal fistula, strabismus, microcephaly, congenital hip dislocation, and Down's syndrome. Considering the significant potential risks, careful monitoring is vital when prescribing alprazolam for pregnant or breastfeeding females are considered. The use of this drug during pregnancy is recommended only when there is no other alternative for the mother and the benefits of using it outweigh the potential risks. This is a very sensitive and important issue in relation to allergies that willow is considered during the first trimester of pregnancy because its use may be associated with an increased risk of congenital anomalies. However, there are still insufficient studies on this drug in pregnant women and in order to know the possible risks associated with the use of this drug on the fetus

(Ait-Daoud et al., 2018; Gidai et al., 2008). Due to reports on the effects of breastfeeding on infants, it may not be a suitable option for repeated use of the benzodiazepine group during lactation. At this point in time, the use of short-acting benzodiazepines without the active metabolite is preferred (Shyken et al., 2019).

Interactions: Since alprazolam is metabolized by the CYP3A4 enzyme, any drug that inhibits the activity of the CYP3A enzymes can disrupt the metabolism of alprazolam, leading to the accumulation of the drug in the body and intensifying its side effects. Many drugs have been reported to have interactions with alprazolam by inhibiting the CYP3A4 enzyme such as erythromycin, cimetidine, norfluoxetine, fluvoxamine, itraconazole, ketoconazole, propoxyphene, nefazodone, and ritonavir (Fleishaker & Hulst, 1994; Greene et al., 1995; Allqvist et al., 2007). For example, simultaneous usage of the imipramine and desipramine with alprazolam increases an average side effect of 31% and 20%, respectively (Schweizer et al., 1993; von Moltke et al., 1995). Concomitant use of alprazolam with alcohol synergizes the effect of the drug on the body, causing enhanced sedative effects, changes in behavior, and intoxication (Bond & Silveira, 1993).



5.4 Long-acting 1,4-benzodiazepines

Long-acting 1,4-benzodiazepines display a longer half-life ($t_{1/2} > 24$ h) with a greater effect on the patient's daily relaxation than other 1,4-benzodiazepines. Upon administration of these drugs, the patient may not feel any improvements on the first night, but on the second or third night. The duration of action for these medications is about 40 to 200 h. However, they contain a risk of accumulation in the elderly and in individuals with severely impaired liver function. On contrary, the severity of rebound effects and withdrawal symptoms of long-acting 1,4-benzodiazepines are lower than those of other types of 1,4-benzodiazepines. Table 5.13 represents the most common FDA-approved long-acting 1,4-benzodiazepine-based drugs along with some details including their brand names, half-life category, and their pharmaceutical applications.

5.4.1 Clorazepate

Introduction: Clorazepate (as a generic name) under the brand name of *Tranxene* (as well as *Tranxene*, *Tranxilium*, *Novo-Clopat*) and with the IUPAC name of 7-chloro-2,3-dihydro-2-oxo-5-phenyl-1

Table 5.13 Overview of FDA-approved long-acting benzodiazepine-based drugs.

Generic name	Brand name	FDA-approved indications	Approval date	Manufacturer
Clorazepate	Tranxene	Anxiety, alcohol withdrawal	1972	Lunbeck
Clonazepam	Klonopin, Rivotril	Anxiety, seizures	1975	Roche
Chlordiazepoxide	Librium	Anxiety, alcohol withdrawal	1960	Valeant
Flurazepam	Dalmane, Dalmadorm	Insomnia	1970	Valeant
Diazepam	Valium, Diastat	Anxiety, alcohol withdrawal, seizures, muscle spasms	1963, 1997	Roche, Valeant
Quazepam	Doral	Insomnia	1985	Questcor

H-1,4-benzodiazepine-3-carboxylic acid, is one of the long-acting 1,4-benzodiazepines widely administered as adjunctive therapy for managing partial seizures and for the treatment of anxiety and acute alcohol withdrawal. An OPPIDUM survey of benzodiazepine uses in France revealed that clorazepate was the second most common benzodiazepine-based drug during the 1990s (Micallef et al., 2016), where clorazepate was mostly used in abuse or dependence situations. Clorazepate is a light-yellow odorless powder that is insoluble in most organic solvents and water. Clorazepate is only stable in its monopotassium or dipotassium salt form, and it has been shown that clorazepate is unstable in aqueous solutions, undergoing rapid decomposition (Tranxene clorazepate dipotassium tablets label, 2010). This instability is originated from the decarboxylation process at position 3 of the diazepine unit, which increases via increasing the solution pH (Clark et al., 1982). Due to the rapid decomposition of clorazepate in the gastrointestinal system to its metabolites, identification of the intact drug in the body fluids is often challenging.

Pharmacodynamics: Clorazepate's mechanism of action is in a similar way to other 1,4-benzodiazepines, activating the frequency of GABA_A receptor and opening chloride (Cl⁻) channels via binding to the benzodiazepine receptors in GABA postsynaptic neurons at several sites in the CNS such as the limbic system and retinal formation. This increase of GABA-mediated chloride influx across the cell membrane leads to hyperpolarization and synaptic inhibition.

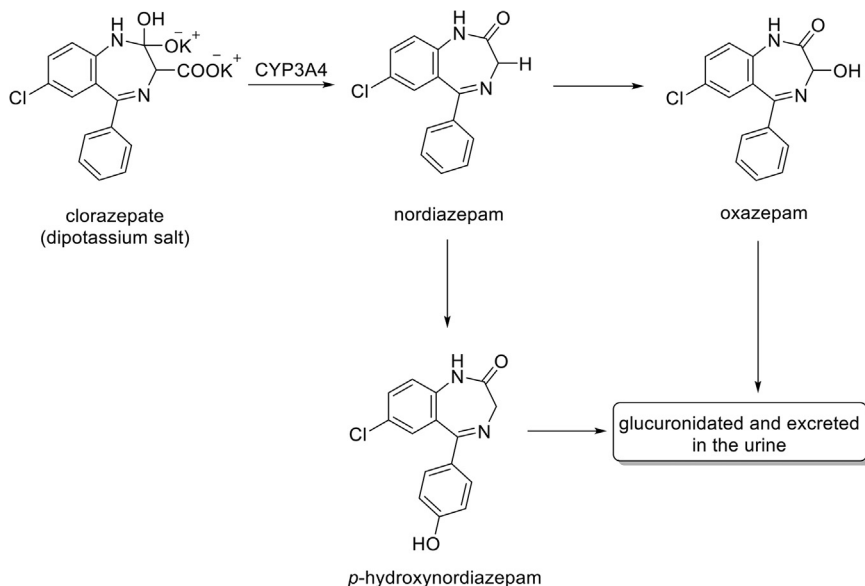


Figure 5.6 Metabolic pathway of clorazepate.

Table 5.14 Pharmacokinetic parameters of clorazepate.

Plasma protein binding	98%
Bioavailability	91%
Volume of distribution (V_d)	0.7–2.2 L/kg
Elimination half-life ($t_{1/2}$)	20–160 h
Time to peak (T_{max})	0.5–2 h
Excretion	Urine (62–67%), feces (15–19%)

Metabolism: Same as other 1,4-benzodiazepines, clorazepate is extensively bound to plasma proteins (98%) and metabolized in the liver, in which about 90% of orally administered clorazepate is excreted in the urine and feces (Fig. 5.6). This drug in the form of dipotassium salt is largely absorbed as *N*-desmethyldiazepam (nordiazepine) upon decarboxylation in the stomach. This major active metabolite, which displays significant anticonvulsant activity at the GABA_A receptor, is further hydroxylated to oxazepam and finally glucuronidated and excreted from the body (Frey & Scherkl, 1988).

Pharmacokinetics and dosage: Although clorazepate has been marketed for decades, little information is available on the pharmacokinetics of this medication. The metabolites of clorazepate reach their peak concentrations within 2 h, which display a long half-life of 20–160 h (Table 5.14)

Table 5.15 Administration profile of clorazepate.

Treatment	Age group	Dosage schedule
Anxiety	Adult	(a) Initial dose: 15 mg, orally at bedtime or in divided doses; (b) Maintenance dose: 15–60 mg orally in divided doses
Alcohol withdrawal syndrome	Adult	Day 1: 30 mg orally followed by 30 to 60 mg in divided doses; Day 2: 45 to 90 mg orally in divided doses; Day 3: 22.5 to 45 mg orally in divided doses; Day 4: 15 to 30 mg orally in divided doses; Day 5: The daily dose should be reduced to 7.5–15 mg (Maximum dose: 90 mg/day)
Seizures	Adult	(a) Initial dose: 7.5 mg orally 3 times a day; (b) Maintenance dose: no more than 7.5 mg orally per week; (c) Maximum dose: 90 mg per day
Seizures	Pediatrics (≥ 12 years)	(a) Initial dose: 7.5 mg orally 3 times a day; (b) Maintenance dose: no more than 7.5 mg orally per week; (c) Maximum dose: 90 mg/day

(Lader et al., 1980; Rey et al., 1979). The recommended initial dosage of clorazepate for the treatment of anxiety, alcohol withdrawal, and seizure in both adults and pediatrics is 15 mg/day, which is administered orally in divided doses. The drug dose should be gradually increased within the range of 15–60 mg in accordance with the response of the patient and with a maximum level of 90 mg per day (Table 5.15). After the patient has been stabilized on a desirable dosage, the frequency of dosing should be reduced to twice or once per day with the main portion taken at night to produce the suitable anxiolytic effect without the appearance of drowsiness or mental impairment (Tranxene clorazepate dipotassium tablets label, 2010). In managing acute alcohol elimination, dosages up to 90 to 120 mg per day are often administered. In the United States and Canada, clorazepate is commercially available as capsules or tablets with 3.75, 7.5, and 15 mg dosages, while tablets marketed in Europe contain 5 mg, 10 mg, 20 mg, and 50 mg of clorazepate.

Adverse effects: No acute toxicity for the long-term single daily doses of clorazepate has been reported. However, patients with the sudden termination of high doses are accompanied by insomnia, nervousness, diarrhea, or mind failure. Clorazepate is not recommended during pregnancy or lactation. The active clorazepate metabolite, nordiazepam, can easily cross the human placenta and is also excreted in breast milk. There is an increased risk of congenital malformations associated with the use of clorazepate during

the first trimester of pregnancy (McElhatton, 1994). Therefore, one should be advised to consult a physician about the termination of the drug if she intends to become pregnant.

Interactions: As clorazepate is often considered as an adjunctive medication in the maintenance treatment of anxiety, its potential drug-drug interaction with the common antidepressant drugs is highly important. In this case, Kiejna and co-workers investigated the effect of clorazepate on the function of a series of neuroleptics and antidepressant drugs in 36 patients for the management of anxiety and aggression (Kiejna et al., 1997). No destructive interactions between the co-administered clorazepate with the other medications were observed, nominating clorazepate as a safe drug with a low risk of drug-drug interactions. On contrary, simultaneous consumption of illicit drugs such as morphine, methadone, opioids, and marijuana (cannabis) with clorazepate can cause severe side effects (Lelong-Boulouard et al., 2006; Gurvich & Cunningham, 2000).

5.4.2 Clonazepam

Introduction: Clonazepam (as a generic name) under the brand names of *Klonopin* and *Rivotril* with the IUPAC name of 5-(2-chlorophenyl)-1,3-dihydro-7-nitro-2H-1,4-benzodiazepine-2-one was initially used as an adjunctive therapy to treat a broad range of primary or secondary generalized seizures (Riss et al., 2008; The US Food and Drug, 2013). It is insoluble in water classified as Class II of the Biopharmaceutical Classification System (BCS), exhibiting high cell-permeability due to its great lipid-solubility (Löbenberg & Amidon, 2000). Due to its slower elimination, clonazepam displays a longer acting than other anti-panic benzodiazepines, e.g., alprazolam and lorazepam, which is why the risk of withdrawal syndromes and rebound anxiety is less in the short-term administration. However, long-term clonazepam use may cause pharmacodynamic tolerance and withdrawal symptoms such as dizziness, headache, seizure, and tremor upon acute termination of clonazepam (Riss et al., 2008). The intensity of these symptoms seems to be affected by several parameters including dosage, the duration of therapy, and the drug-tapering rate. Clonazepam is also effective for treating generalized epilepsies in patients with no seizures. Dreifuss et al. made an analysis on 10 epileptic children with absence seizures treated with clonazepam, where 7 patients had a 75% decrease in seizure frequency after 8 weeks of the drug administration (Dreifuss et al., 1975). In another clinical study, it has been shown that clonazepam is the drug of choice in rare childhood epilepsy called Northern epilepsy syndrome (Hirvasniemi et al., 1995).

Pharmacodynamics: Although the exact mechanism of action of clonazepam has not been well understood, it is suggested that this medication may increase the level of serotonin in the brain and also enhance the concentration of the neurotransmitter at synaptic receptor sites (Davidson & Moroz, 1998). In a preliminary study carried out by Battistin and co-workers (Battistin et al., 1984), the effect of clonazepam was investigated on brain GABA levels, glutamate decarboxylase (GAD), and GABA_T activities. The drug exhibited inhibition of GAD activity but had no direct effects on GABA_T and GABA levels. In general, the primary action sites of benzodiazepines are located at the GABAergic synapses, where they increase GABA transmission. Binding benzodiazepines to GABA receptors is associated with changes in permeability to ions, modulating neuronal activity. In this case, the GAD inhibition induced by clonazepam imitates this process in the human brain. Since GABA affects prolactin secretion in the hypothalamus, the potency of three 1,4-benzodiazepines, e.g., clonazepam, diazepam, and chlordiazepoxide, was investigated to inhibit prolactin release correlated to GABA. While the inhibitory activities of diazepam and chlordiazepoxide on the prolactin release have been found to be dose-dependent, all analyzed doses of clonazepam are effective and this medication displays the highest potency compared with the other drugs (Grandison, 1982).

Metabolism: Clonazepam with 86% plasma protein binding is majorly metabolized in the liver by CYP3A4 and CYP3A4 enzymes through the reduction of the nitro group to form the 7-amino clonazepam, which is subsequently converted to 7-acetamido clonazepam by N-acetyl transferase 2 (NAT2) (Fig. 5.7). Hydroxylation of the parent molecule and the resulting metabolites at the C3 position as the minor metabolic pathway also occurs to give the corresponding 3-hydroxy metabolites (Tokola & Neuvonen, 1983; Tóth et al., 2016). 7-Amino clonazepam species has the highest concentration in the plasma compared to the other metabolites, even though it is recognized to be pharmacologically inactive. Nevertheless, an *in vivo* study on mice demonstrated that the 7-amino metabolite could compete with the parent drug to occupy benzodiazepine-binding sites, leading to a quick reduction of clonazepam attachment to the GABA receptors after the termination of treatment (Munakata & Tsuchiya, 2008).

Pharmacokinetics and dosage: Clonazepam can be administered through intravenous (i.v.), intramuscular (i.m.), oral, and rectal routes in the treatment of epileptic patients and panic disorders (Berlin & Dahlström, 1975; Browne, 1976). It is highly absorbed after i.m. and oral administration, displaying an average bioavailability of 93% and 90%, respectively. Oral

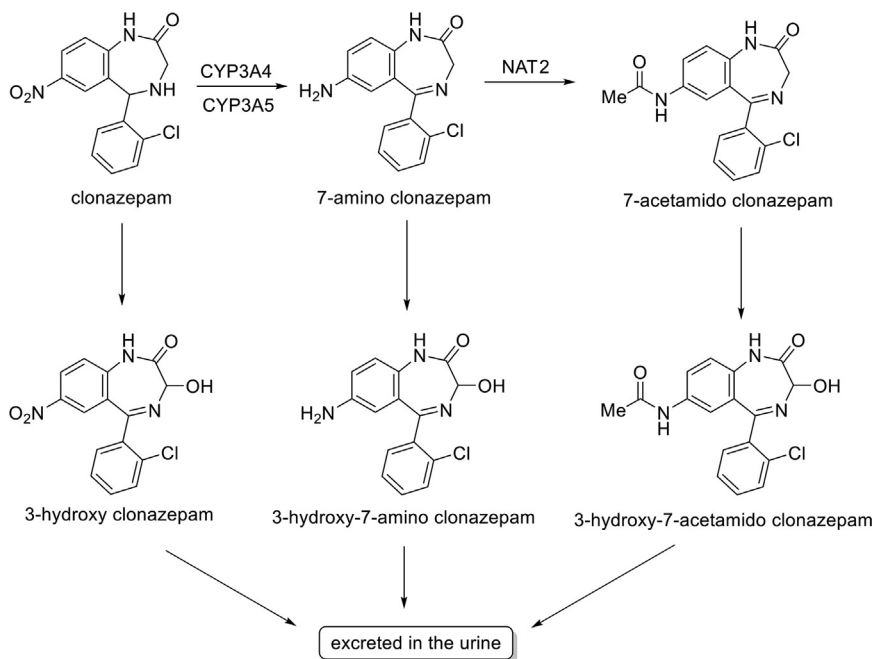


Figure 5.7 Metabolic pathway of clonazepam.

clonazepam therapy is the method of choice for patients with panic disorders, displaying fewer withdrawal-related symptoms and a longer duration of action compared with other anti-panic medications such as alprazolam and lorazepam (Davidson, 1997). The i.m. administration of clonazepam has no privilege over its oral route, which is often prescribed at the same dose as an oral formulation. However, i.m. method may be useful for patients with dysphagia and intense seizures, where the i.v. route is not possible. In the case of prolonged seizures, intravenously administration of anticonvulsants, such as diazepam and clonazepam, is an effective therapy. However, some situations may occur where these drugs cannot be prescribed intravenously. In these cases, rectal administration could be a promising alternative (Klosterskov Jensen et al., 1983). Diazepam is the common medication to be used in this way, which causes respiratory depression. On contrary, clonazepam exhibits much fewer side effects than diazepam, nominating it as a suitable drug for continuing convulsions. The recommended initial dosages of clonazepam for the treatment of seizures and panic disorders in adults are 0.5 mg/day and 0.25 mg/day, respectively, with a maximum dose limit of 20 mg daily. A significantly reduced amount of 0.01 mg/kg/day clonazepam is often

Table 5.16 Administration profile of clonazepam.

Treatment case	Adults (< 65 years old)	Pediatrics (≤ 10 years old)
Seizures	0.5–20.0 mg/day	0.01–0.05 mg/kg/day
Panic disorders	0.25–4.0 mg/day	–

Table 5.17 Pharmacokinetic parameters of clonazepam after single dose of 2 mg.

Administration type	$C_{\max}$ (ng/mL)	$T_{\max}$ (h)	AUG (ng/mL $\times$ h)	$t_{1/2}$ (h)
Intravenous (i.v.)	27.0	< 0.1	613	38
Intramuscular (i.m.)	11.0	3.1	620	43.6
Oral	14.9	1.7	561	39.0
Rectal ^a	18.4–40.5	0.1–2	NR	NR

^aAdministered 0.1 mg/kg of body weight. (NR = not reported).

administered for pediatrics with seizures due to their low CYP3A-catalyzed drug metabolism (Table 5.16) (Browne, 1976; The US Food and Drug, 2013).

A clinical study for the investigation of the pharmacokinetics of clonazepam was conducted with 12 healthy volunteers who were given a single dose of 2 mg clonazepam through i.v., i.m., and oral routes (Table 5.17) (Crevoisier et al., 2003). The absorption rates after i.m. and oral prescriptions are considerably different, as the plasma concentration reaches the maximum of 11.0 and 14.9 ng/mL after 3.1 h and 1.7 h, respectively. The elimination half-life of clonazepam can vary from 38 to 44 h based on the administration type. This long half-life may decrease the severity of clonazepam withdrawal syndrome compared to other benzodiazepine drugs. No significant difference is observed between the area under the plasma concentration–time curves (AUC) of the i.m. and oral administration, indicating that almost the same amount of drug comes into the systemic circulation after either i.m. or oral routes. In the case of the rectal route, plasma concentration peaks ranging 18.4–40.5 ng/mL occur in 0.1–2 h after given 0.1 mg/kg clonazepam (Table 5.17) (Rylance et al., 1986).

Adverse effects: If patients are treated with a low dose of clonazepam (< 2.0 mg/day), mild to moderate adverse reactions may occur. A clinical study on 69 patients with anxiety disorder compared the efficiency of clonazepam with lorazepam and alprazolam (Wang et al., 2016), in which the individuals were administered orally with 0.25, 0.5, and 1 mg/day of clonazepam, lorazepam, and alprazolam, respectively, for a period of 6 weeks. The results showed that all three drugs are equally potent in the

treatment of anxiety. The incidence of adverse events is notably lower in the clonazepam-administrated patients than with the other two medications (clonazepam = 26.7%, alprazolam = 48.4%, and lorazepam = 43.9%), where somnolence (9.3%) and dizziness (6.7%) are the most common adverse reactions for clonazepam. According to the FDA guideline for clonazepam use, 50% of people suffering from seizure disorders treated with clonazepam experience drowsiness ([The US Food and Drug, 2013](#)). As mentioned earlier, there are no severe withdrawal symptoms in the short-term administration of clonazepam, as somnolence and depression are the most common adverse events. No comprehensive clinical study is available regarding the risks of clonazepam taken by pregnant women to their unborn babies. A few surveys have reported congenital anomalies, paralytic ileus, and apnea among the infants of mothers with epilepsy who used clonazepam during their pregnancy ([Fisher et al., 1985](#); [Czeizel et al., 1992](#); [Haeusler et al., 1995](#)). Taking these adverse effects into mind, clonazepam is recommended to be prescribed during pregnancy only if its clinical benefits to the mother outweigh the risks to the fetus, though the use of the lowest dose for the shortest time is not in the first trimester. Like other benzodiazepines, clonazepam is excreted in human breast milk. Infants exposed to the drug during breastfeeding should be monitored for CNS depression or apnea ([The US Food and Drug, 2013](#)). Therefore, extreme caution is advised when clonazepam is administered during lactation.

Interactions: Clonazepam is mainly cleared via oxidative metabolism; therefore, it is susceptible to have interactions with those drugs that modulate oxidative metabolism such as amiodarone ([Witt et al., 1993](#)). Furthermore, an interaction may occur between clonazepam and phenytoin (a famous antiepileptic drug), in which the addition of clonazepam significantly reduces phenytoin plasma concentration levels from 24.8 to 16 $\mu\text{g/ml}$, even at high dosages of phenytoin ([Saavedra et al., 1985](#)). Co-consumption of clonazepam and alcohol causes increased drowsiness and depressed breathing as these substances enhance intoxicating effects of each other.

5.4.3 Chlordiazepoxide

Introduction: Chlordiazepoxide (as a generic name) with the IU-PAC name of 7-chloro-2-methylamino-5-phenyl-3*H*-1,4-benzodiazepin-4-oxide is the first benzodiazepine compound accidentally discovered by Leo Sternbach in 1955 when he was working on the development of tranquilizers ([Sternbach, 1972](#)). The first clinical studies were then conducted on schizophrenic patients, where chlordiazepoxide was found to exhibit no

specific antipsychotic effect. Instead, it significantly reduced anxiety in the patients, signifying its potential as an antianxiety agent. Initial animal testing also showed that chlordiazepoxide had substantial sedative, anticonvulsant, and muscle relaxant effects. In 1960, three extensive clinical trials of chlordiazepoxide were carried out on patients with psychoneurotic disorders and chronic depression. Results showed the noticeable therapeutic efficiency of chlordiazepoxide with minor adverse effects on intellectual activity (Harris, 1960). Soon after, Hoffmann-La Roche commercialized it as “*Librium*”, which was one of the major revolutions in the pharmacological treatment of anxiety disorders. Prior to the discovery of chlordiazepoxide, meprobamate and barbiturates were the drugs of choice to treat patients with serious anxiety conditions (Wick, 2013). However, these compounds had severe physical dependence and potential withdrawal syndrome. Chlordiazepoxide has also been tested on mice, rats, cats, and dogs, displaying an excellent muscle relaxation effect (López-Muñoz et al., 2011). Chlordiazepoxide is a weak base ($pK_a = 4.8$) with poor aqueous solubility at physiological pH. Therefore, this drug is often formulated in the form of hydrochloride salt (prepared at acidic pH of around 3.0) to ensure its water solubility in the body. The configuration of the nitrone group in chlordiazepoxide makes it unstable in the presence of ultraviolet light or in solution (L. Sternbach et al., 1962). Accordingly, oral dosage forms are supplied in opaque capsules or the drug solution is freshly prepared before use.

Pharmacodynamics: The mechanism of action of chlordiazepoxide is not completely known, however, it is suggested that this medication works through opening the chloride channel located on the GABA receptor, which consequently induces the chlorine ion influx inside the cell membrane and results in hyperpolarization. This phenomenon accordingly has a sedative and anxiolytic effect on the central nervous system (Choi et al., 1977).

Metabolism: The metabolism of chlordiazepoxide in humans is a complex process that consists of biotransformation into several long-acting pharmacologically active metabolites including desoxychlordiazepoxide, demoxepam, desmethylchlordiazepoxide (norchlordiazepoxide), desmethyldiazepam (nordiazepam), and oxazepam (Fig. 5.8) (Greenblatt, Shader, MacLeod, & Sellers, 1978). The drug is first metabolized to norchlordiazepoxide and desoxychlordiazepoxide catalyzed by the CYP3A4 enzyme in the liver. Norchlordiazepoxide is then biotransformed to demoxepam, followed by degradation to form nordiazepam, which is also produced from the conversion of desoxychlordiazepoxide (Greenblatt, Shader, MacLeod, Sellers, et al., 1978) (Greenblatt, Shader, Franke, et al., 1978). Note that the

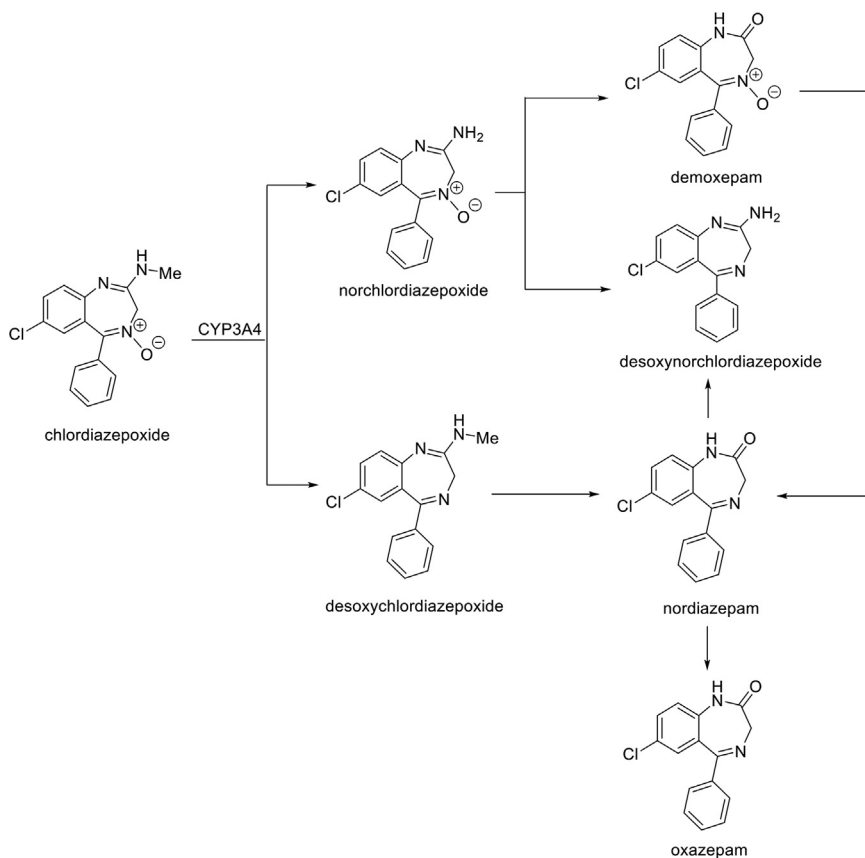


Figure 5.8 Metabolic pathway of chlordiazepoxide.

nordiazepam metabolite can be oxidized to an active oxazepam metabolite. Eventually, all metabolites are conjugated with glucuronide and excreted in the urine.

Pharmacokinetics and dosage: Chlordiazepoxide is pharmaceutically available in opaque capsules with 5 mg, 10 mg, or 25 mg chlordiazepoxide hydrochloride salt. Clinical studies demonstrated that the pharmacokinetics of chlordiazepoxide depend on the administration type (Greenblatt, Shader, MacLeod, Sellers, et al., 1978; Greenblatt, Shader, Franke, et al., 1978; Schwartz et al., 1971). Pharmacokinetics parameters of the medication following three i.v., i.m., and oral administration are compared in Table 5.18. The i.m. chlordiazepoxide is adsorbed more slowly than oral administration due to faster gastrointestinal absorption. There is no significant difference observed between the volume distributions of the three routes, indicating

Table 5.18 Pharmacokinetic parameters of chlordiazepoxide after single dose.

Administration type	C _{max} (µg/mL)	V _d (L/kg)	T _{max} (h)	t _{1/2} (h) ^a
Intravenous (i.v.) ^b	NR	0.26–0.58	NR	5–30
Intramuscular (i.m.) ^c	0.48–1.23	0.25–0.50	2.5–12	5–30
Oral ^c	1.15–2.07	0.25–0.50	1–12	5–30

^a Elimination half-life of active metabolites = 36–200 h. (NR = not reported).

^b 30 mg dosage.

^c 50 mg dosage.

that almost the same amount of drug is distributed into the systemic circulation. Although the parent compound displays a 5–30 h elimination half-life ($t_{1/2}$), its corresponding active metabolites have $t_{1/2} = 36–200$ h. Therefore, chlordiazepoxide is considered as a long-acting 1,4-benzodiazepine prescribed for long-term treatment. Similar to other benzodiazepines, the aging process has a significant effect on the pharmacokinetics of chlordiazepoxide. In a clinical study conducted by Shader and co-workers (Shader et al., 1977), a number of healthy individuals ranging from 30–60 years old were given a single 25 mg oral dose of chlordiazepoxide. The results showed that the average elimination half-life in the elderly cases was remarkably longer than that of the young volunteers. Furthermore, the total clearance of the drug was significantly reduced in the elderly. These observations may be due to the increased sensitivity of aged people to the CNS depressant effects.

The recommended dosages of chlordiazepoxide for the treatment of anxiety or acute alcohol withdrawal syndrome highly depend on the patient's response to the drug. In the case of mild and moderate anxiety disorders in adults, the initial oral dose is 5 mg administered 3–4 times daily, with a maximum dose limit of 300 mg/day (The US Food and Drug, 2005). People with severe anxiety symptoms are recommended to take 20 mg chlordiazepoxide orally, 3 to 4 times daily. The usual starting dose of the drug for geriatrics is also 5 mg given 2–4 times per day. Chlordiazepoxide is one of the best medications for acute alcohol withdrawal therapy, in which the initial dose of 50 to 100 mg is suggested, followed by repeated doses as required until agitation is managed, with no more than 300 mg daily (The US Food and Drug, 2005).

Adverse effects: The occurrence and severity of the adverse reactions caused by the chlordiazepoxide were much less than those of other benzodiazepines. For example, ataxia, drowsiness, and confusion are some of the minor side effects reported in aged patients (The US Food and Drug, 2005). Agranulocytosis, jaundice, and hepatic dysfunction have also been observed during chlordiazepoxide drug therapy. In this case, periodic blood counts

and liver function analyses are highly recommended, since chlordiazepoxide treatment is protracted ([The US Food and Drug, 2005](#)). Chlordiazepoxide is categorized by the Drug Enforcement Administration (DEA) as a Schedule IV controlled drug, representing the least potential for abuse and low risk of dependence ([The US Food and Drug, 2005](#)). The intensity of withdrawal symptoms because of abrupt termination of chlordiazepoxide highly depends on the drug dose and the therapy. The more excessive drug doses over an extended time, the more severe withdrawal events can occur. Therefore, after extended treatment, a gradual dosage tapering schedule should be followed. There is an increased incidence of congenital malformations to the human fetus as a result of the use of chlordiazepoxide during the first trimester of pregnancy ([Czeizel et al., 2004](#)). Since the drug can be excreted into human milk, it is also recommended not to use chlordiazepoxide by mothers during breastfeeding.

Interactions: The coadministration of chlordiazepoxide and opioids may increase the risk of respiratory depression due to disruptive actions at respiratory-related receptors in the CNS ([The US Food and Drug, 2005](#)). Furthermore, there is a high risk of potential drug interactions associated with the concomitant use of chlordiazepoxide with other benzodiazepines, antidepressant agents, and antihistamines such as alprazolam, lorazepam, zolpidem, fluoxetine, fluvoxamine, nefazodone, cimetidine, clozapine, cetirizine, and diphenhydramine ([Kurowski & Reim, 1986](#); [Feldman & Smith, 1978](#); [Rodgers et al., 1997](#)). Alcohol and marijuana (cannabis) can also intensify the adverse effects of chlordiazepoxide such as dizziness, drowsiness, impairment in thinking and concentrating ([Chan et al., 1982](#)).

5.4.4 Flurazepam

Introduction: Flurazepam (as a generic name) under the brand name of *Dalmane* or *Dalmadorm* and IUPAC name of 7-chloro-1-(2-(diethylamino)ethyl)-5-(*o*-fluorophenyl)-1,3-dihydro-2*H*-1,4-benzodiazepine-2-one is a long-acting 1,4-benzodiazepine known as a hypnotic drug for the treatment of insomniacs with difficulty in falling asleep, frequent nocturnal awakenings, and early morning awakenings ([The US Food and Drug, 2007](#)). It was initially introduced to the US as a sleep-inducing agent in 1970 and soon later became popular reaching 6.5 million prescriptions in 1973. Flurazepam is available as a yellow crystalline dihydrochloride salt that is highly soluble in water and alcohol. This medication is more potent than a placebo and displays the same efficiency as other hypnotic drugs in short-term treatments.

However, flurazepam is more effective in long-term dosage therapy than other hypnotics such as barbiturates, glutethimide, chloral hydrate, meprobamate, and methaqualone (Greenblatt et al., 1975a). These medications become ineffective when administered constantly over two weeks, however, flurazepam persists to provide pharmacologically effects even when ingested repeatedly for up to 4 weeks (Kales et al., 1970). These features signify flurazepam's advantages over other nonbenzodiazepine hypnotics.

Pharmacodynamics: Several studies have described that flurazepam displays more central CNS depression and less anxiety reduction than other benzodiazepines (Randall et al., 1969; Tseng & Wang, 1971), which clarifies flurazepam's usage as a hypnotic rather than as an antianxiety drug. These observations demonstrate that flurazepam acts as a central muscle relaxant via blockade of GABAergic interneurons at supraspinal level with no effects on the spinal cord. Cannizzaro and co-workers investigated the effect of flurazepam on rats after single-dose administration, where the medication resulted in CNS depression but no behavioral disinhibition (Cannizzaro et al., 1972). However, the mice showed tolerance to the depressant effects of flurazepam upon repeated ingestion, followed by the appearance of behavioral disinhibition. The drug has also an inhibitory effect on the direct electrical stimulation of interneurons in the brainstem reticular system, exhibiting no impact on both monosynaptic and polysynaptic reflex behaviors (Greenblatt et al., 1975a). In a study on mice given ataxia-producing doses of several benzodiazepines (Fennessy & Lee, 1972), it was shown that flurazepam decreases the amount of dopamine in their brain, whereas other benzodiazepines including diazepam, nitrazepam, chlordiazepoxide, clonazepam, and medazepam represent either no change or increased the dopamine brain content. Although many studies have been carried out on flurazepam behavior, the clinical connection of these analyses is not known yet.

Metabolism: Flurazepam concentration in the blood after administration is extremely low, which is observable only for a few hours after a single dose (Miller et al., 1988). The metabolic pathway of flurazepam is represented in Fig. 5.9, consisting of the formation of three metabolites. Flurazepam is quickly absorbed from the GI tract (within 30 min) and biotransformed to its metabolic derivatives. Concentrations of two major metabolites, e.g., hydroxyethyl flurazepam and desalkylflurazepam, become significantly high after approximately one hour of administration, where the desalkylflurazepam metabolite stays in the blood for a longer period. Therefore, these pharmacologically active metabolites may cause flurazepam's

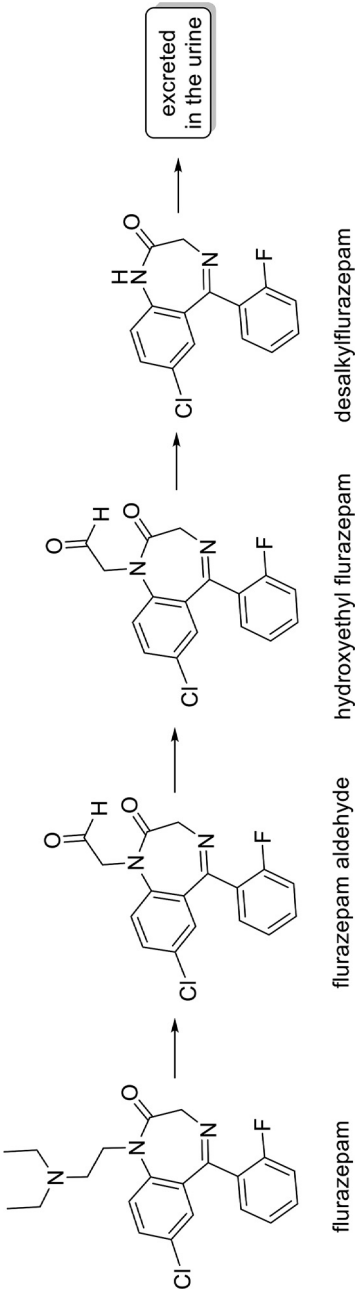


Figure 5.9 Metabolic pathway of flurazepam.

Table 5.19 Pharmacokinetic parameters of desalkylflurazepam as the major flurazepam metabolite after single 15 mg oral dose.

Gender (age)	Weight (kg)	C _{max} (ng/mL)	T _{max} (h)	t _{1/2} (h)
Male (19–34)	62–86	5.3–9.6	0.5–16	37–118
Female (20–33)	52–61	11.5–20.0	0.75–73	35–133
Male (66–72)	66–89	8.2–15.1	0.75–96	71–289
Female (61–85)	46–73	10.3–19.1	0.75–24	85–165

Table 5.20 Pharmacokinetic parameters of flurazepam and its metabolites after single 30 mg oral dose.

Species	C _{max} (mean) (ng/mL)	T _{max} (mean) (h)	Total AUC (mean) (ng/ml × h)	t _{1/2} (mean) (h)
Flurazepam	0–10 (2.4)	0.5–2.0 (1.0)	0.5–4.4 (1.8)	0–19.0 (5.3)
Flurazepam aldehyde	3.1–19.3 (7.8)	0.75–2.5 (1.3)	1.3–28.9 (6.9)	7.4–155.7 (48.3)
Hydroxyethyl flurazepam	7.1–23.4 (15.1)	0.75–2.0 (1.1)	0.9–2.4 (1.3)	14.8–65.2 (38.4)
Desalkyl flurazepam	7.4–47.0 (20.4)	1.0–72 (10.2)	34.1–154 (71.4)	911–3084 (1969)

clinical effects in the human body (Miller *et al.*, 1988). Flurazepam and its metabolic derivatives are finally excreted in the urine, where hydroxyethyl flurazepam and desalkylflurazepam consist of 22%–55% and 1% of the dose, respectively.

Pharmacokinetics and dosage: In a clinical study, Greenblatt *et al.* have made an analysis on the pharmacokinetics of desalkylflurazepam as the major metabolite in 26 healthy noninsomniac volunteers aging from 19 to 85 years old after orally given 15 mg single dose of flurazepam (Greenblatt *et al.*, 1981). Maximum desalkylflurazepam plasma concentration ranging from 5.0 to 20.2 ng/mL occurs between 0.5 and 96 h after administration, which is found to be higher in females possibly due to their lower body weights (Table 5.19). The elimination half-life of the metabolite increases with age in both genders as changes in body composition with age may influence the drug uptake of tissues, leading to altered pharmacokinetics. In a similar clinical analysis, the relative contribution of the parent drug and its metabolites in the human blood was investigated (Table 5.20) (Miller *et al.*, 1988). The mean maximum plasma levels of flurazepam and flurazepam aldehydes are 2.4 and 7.8 ng/mL, respectively, which become undetectable within the respective 3 h and 6.9 h after administration. Hydroxyethyl

flurazepam reaches the average concentration level of 15 ng/mL after 1.1 h, becoming unobservable after 8 h. On the contrary, desalkylflurazepam shows the highest plasma peak concentration (20.4 ng/mL) that occurs at 10.2 h after dosing with an average $t_{1/2} = 71.4$ h. This metabolite remains in the blood much longer than other species, which is still detectable after 9 days of administration.

Flurazepam is marketed as capsules containing 15 mg or 30 mg flurazepam hydrochloride (The US Food and Drug, 2007). The recommended dosage for adults is 30 mg before sleeping at night. However, this drug amount is reduced to 15 mg for elderly people (> 65 years old). There have been no clinical analysis of flurazepam in children, which is why this medication is not recommended for use in people under 15 years old.

Adverse effects: Flurazepam has an advantage over other hypnotics as there is no rapid eye movement (REM) rebound observed upon discontinuation of the drug (Isjanovski & Isjanovski, 2019). According to the FDA flurazepam instruction label, hangover, drowsiness, headache, and dry mouth are the most common side effects of flurazepam (The US Food and Drug, 2007). Studies have also shown that this drug has significantly less potential for abuse and displays less toxicity when ingested in excess (Greenblatt et al., 1975a; Greenblatt et al., 1975b). Flurazepam decreases the sleep time in stages 3 and 4, being maintained drug the medication withdrawal (Kales et al., 1971). Flurazepam is categorized as a Class X by the FDA, indicating that it should not be used during pregnancy (The US Food and Drug, 2007). Despite the global extensive administration of flurazepam, there is little clinical information available about its effect on the fetus during pregnancy. A study on the effect of flurazepam during organogenesis of the rat fetus has described that the medication ingestion during the second half of pregnancy may result in severe physical abnormalities (Takzare et al., 2008). The FDA has also reported the presence of high plasma concentration of desalkylflurazepam in the infant due to the transplacental transfer of this long-acting metabolite to the fetal circulation (The US Food and Drug, 2007).

Interactions: Common nonbenzodiazepine hypnotics such as barbiturates and glutethimide are known to modulate the performance of hepatic microsomal enzymes, causing severe interactions with other drugs, especially in the case of oral anticoagulant drugs. In contrast, flurazepam exhibits no significant enzyme induction, where it can be prescribed to patients simultaneously taking oral anticoagulants (Greenblatt et al., 1975b). Several studies have also shown that there is no noticeable interaction of small

doses of flurazepam with alcohol in the human (Greenblatt et al., 1975b). Nevertheless, patients should be advised that flurazepam in high dosages may increase the CNS depression resulted from alcohol and other CNS-depressant drugs.

5.4.5 Diazepam

Introduction: Diazepam (as a generic name) under the brand name of *Valium* and IUPAC name of 7-chloro-1,3-dihydro-1-methyl-5-phenyl-2H-1,4-benzodiazepin-2-one is the second clinically approved benzodiazepine widely used for the treatment of intensive care sedation, repeated seizures, and short-term anxiety (Dundee & Haslett, 1970; The US Food and Drug Administration, 2016c). Following the discovery of chlordiazepoxide, the first marketed benzodiazepine introduced in 1957, further structural alterations on this compound were carried out to improve its biological performance, leading to the synthesis of diazepam as a more potent benzodiazepine. This drug was first introduced to the market by Hoffman-La Roche in 1963, and soon later it became the top-selling anxiolytic in the US from 1968 to 1982, which may owe to its quick onset of action and high potency with low toxicity (L. H. Sternbach, 1978). This colorless crystalline compound is insoluble in water with a calculated partition coefficient ($cLogP$) and topological polar surface area (TPSA) of 3 and 32.7, respectively, indicating good drug metabolism behavior and CNS penetration (Calcatterra & Barrow, 2014). Due to the well-characterized effects of diazepam in the literature, most laboratory analyses of anxiety on animals often employ diazepam as a control test to establish the validity of the assay (Calcatterra & Barrow, 2014).

Pharmacodynamics: Diazepam is known as a positive allosteric modulator of the GABA_A receptor complex, which attaches to specific binding sites on the α - γ subunit interfaces. Upon the interaction of the drug with these sites, chloride ion influx in the neurons significantly boosts that leads to hyperpolarized postsynaptic membranes and subsequently increasing CNS depression (Costa et al., 1978). This phenomenon eventually has anxiolytic and antiepileptic effects on neuronal processes in the thalamus, hypothalamus, limbic system, and cerebral cortex. In addition to this mechanism of action, some studies have shown the possibility of diazepam binding to voltage-gated sodium and calcium channels that may be associated with its anticonvulsant effects (Willow et al., 1984; Taft & DeLorenzo, 1984). Diazepam is also found to block stress-induced prolactin release (Grandison, 1982), acetylcholine release (Miller & Richter, 1985), and histamine recycling in mice brains (Oishi et al., 1986). Regardless of these various mechanisms

of action, the majority of the diazepam pharmacological activities seem to be mediated via the positive allosteric modulatory effects of diazepam on the GABA_A chloride channels.

Metabolism: Diazepam is metabolised in the liver by CYP450 enzymes and biotransformed to three main active metabolites, desmethyldiazepam (nordiazepam), temazepam, and oxazepam (Fig. 5.10) (Hillestad, Hansen, & Melsom, 1974; Hillestad, Hansen, Melsom, et al., 1974). Initially, the parent compound is demethylated by CYP2C9, CYP2C19, CYP2B6, CYP3A4, and CYP3A5 to give nordiazepam. In an alternative route, diazepam is also 3'-hydroxylated by CYP3A4 and CYP3A5 enzymes to form temazepam. These metabolites are further hydroxylated or demethylated to afford oxazepam, which is finally glucuronidated and excreted by the urinary system. Typically, a trace amount of intact diazepam can be found in the urine along with approximately the whole original drug dose that is eliminated as metabolites. Nordiazepam constitutes the majority of the circulating dose during daily administration and remains much longer than the other metabolites in the blood, reaching steady-state after three weeks.

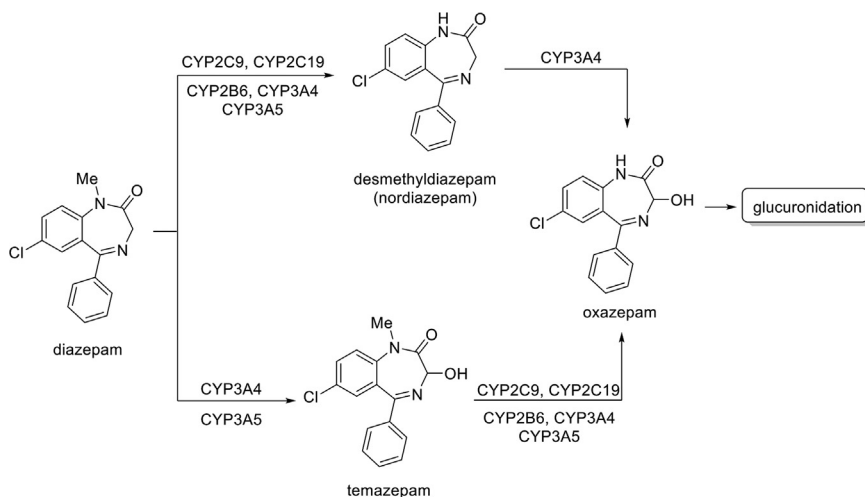


Figure 5.10 Metabolic pathway of diazepam.

Pharmacokinetics and dosage: Diazepam is commercially available as tablets (2 mg, 5 mg, or 10 mg), syrup (5 mg/ml), and ampoules (5 mg/ml). All three diazepam active metabolites have been pharmaceutically marketed as anxiolytics and anticonvulsants due to their potent activity. Temazepam has a short elimination half-life ($t_{1/2} = 8\text{--}20$ h) that nominates it for the treatment of insomnia. Oxazepam displays a slow onset of action (~ 60 min) and no

Table 5.21 Pharmacokinetic parameters of diazepam after single 10 mg dose.

Administration type	C _{max} (ng/mL)	V _d (L/kg)	T _{max} (h)	t _{1/2} (h)
Oral	177	0.8–1.0	90	24–48
Intramuscular (i.m.)	100 ^a , 149 ^b	1.1–2.0	90	24–48
Intravenous (i.v.)	400 ^c	1.1–2.0	15	24–48
Rectal	369	1.1–2.0	17	24–48

^a Injection in buttock.^b Injection in thigh.^c Reached 500 ng/mL with 20 mg dose.

hepatic oxidation is required for elimination, making it suitable for elderly patients or those with hepatic impairment suffering from anxiety (Calcaterra & Barrow, 2014). Diazepam is a lipid-soluble compound widely distributed across the central nervous system ($V_d = 0.8\text{--}2.0$ L/kg) (Table 5.21). It is extensively bound to plasma protein (96–99%) with a high bioavailability of 90–100%. After a single oral administration, the drug is completely absorbed from the GI tract and peak plasma levels reach 65–145 ng/mL within 30–90 min (Table 5.21). Age is the key factor to affect the drug plasma concentration, in which children and elderly people display the highest and lowest values (Mandelli et al., 1978). Intramuscular administration has a poorer absorption rate and peak plasma levels occur after 90 min of injection. More importantly, the concentration levels are higher in an injection in the thigh, possibly due to a richer blood supply to the thigh than the buttock (Table 5.21). In the emergency control of seizures, diazepam is the drug of choice, where its anticonvulsant activity usually appears at peak plasma levels of 400–500 ng/mL through intravenous injection of 10–20 mg. However, if the intravenous route cannot be achieved in elderly patients with intensive seizures, the best alternative is the rectal administration of diazepam (Seigler, 1990). The drug elimination half-life normally occurs in 24–48 h, increasing in pediatric patients due to decreased albumin levels in their serum.

The initially recommended dosages for the management of anxiety disorders in adults are from 2 to 10 mg administered orally 2–4 times daily (The US Food and Drug Administration, 2016c), with a maximum dose of 30 mg. For the relief of acute alcohol withdrawal symptoms, 10 mg of diazepam is usually required taken orally 3–4 times during the first 24 h, followed by a reduction to 5 mg as needed. Muscle antispasm uses typically start from 2 mg, adjunctively administrated orally 3 or 4 times per day. In geriatric patients, it is highly suggested that the drug dosage should be limited to the smallest amount (2–2.5 mg once or twice daily) in order to prevent the development of oversedation, which can be then

increased gradually as needed ([The US Food and Drug Administration, 2016c](#)). All four administration routes of diazepam (oral, i.m., i.v., or rectal) are also used adjunctively with other antiseizure medications in the treatment of convulsive disorders, 2–10 mg, 2 to 4 times daily. Diazepam is strictly prohibited for use in pediatric patients under 6 months. Due to various responses to CNS-acting drugs in pediatric patients (< 18 years old), the therapy should be initiated with the lowest doses of 1 mg to 2.5 mg, 3–4 times daily ([The US Food and Drug Administration, 2016c](#)).

Adverse effects: Fatal adverse reactions related to diazepam such as respiratory arrest and prolonged seizures are very rare, occurring due to prolonged habitual use and the consequence interaction with other anxiolytics, e.g., opioids or alcohol. The most commonly reported minor side effects of diazepam are fatigue, drowsiness, ataxia, and muscle weakness ([The US Food and Drug Administration, 2016c](#)). Diazepam has a lower potential risk for dependence and abuse than other benzodiazepines, which is why it is commonly prescribed for the treatment of withdrawal reactions from alcohol or other benzodiazepines. Nevertheless, acute discontinuation of the long-term treatment may be associated with withdrawal symptoms such as abdominal and muscle cramps, headache, anxiety, tension, confusion, and irritability. Diazepam is subject to Schedule IV control under the Controlled Substances Act of 1970 ([The US Food and Drug Administration, 2016c](#)). The drug doses can be gradually tapered off before a complete termination, to preclude the occurrence of dependence and withdrawal effects from chronic prescription, dosages can be slowly tapered off before complete termination. Diazepam is classified in Category D pregnancy drugs, indicating that it has some human fetus risks based on side effects and the drug can be administered during pregnancy if its potential benefits outweigh its potential risks. The drug and its metabolites can easily cross the placenta and penetrate the fetus's systematic circulation. The teratogenic risk for embryo or fetus in utero, orofacial clefts, cardiovascular malformations are the most common adverse effects of diazepam administered in high doses during pregnancy ([Erkkola et al., 1974](#)). Nevertheless, it has been shown that the short-term drug treatment in usual doses during pregnancy leads to no noticeable teratogenic risk to the fetus ([Erkkola et al., 1974](#)).

Interactions: The concomitant use of diazepam with centrally acting agents such as phenothiazines, barbiturates, antipsychotics, and anticonvulsants should be carefully monitored to prevent any potential interactions ([The US Food and Drug Administration, 2016c](#)). Simultaneous use of diazepam with alcohol is not advised because of escalating the sedative

effect. Furthermore, compounds that disturb hepatic enzymes function, e.g., ketoconazole, cimetidine, fluoxetine, fluvoxamine, and omeprazole, can affect the pharmacokinetics of diazepam, leading to prolonged sedation. Drinking grape juice should be avoided with diazepam as it may interrupt the liver's ability to properly metabolize the drug and allows it to remain in the body longer (Ozdemir et al., 1998; Greenblatt, Allen, et al., 1978).

5.4.6 Quazepam

Introduction: Quazepam (as a generic name) under the brand name of *Doral* with the IUPAC name of 7-chloro-1-(2,2,2-trifluoroethyl)-5-(*o*-fluorophenyl)-1,3-dihydro-2*H*-1,4-benzodiazepin-2-thione is known as a long-acting 1,4-benzodiazepine hypnotic, which has been demonstrated to be highly efficient in inducing sleep in both healthy people and insomniacs (Freemon et al., 1977; Roth et al., 1979). It is a white crystalline compound having some structural similarities with flurazepam and temazepam, which is soluble in ethanol but with poor solubility in water. Quazepam improves both sleep efficacy and maintenance through a minimal effect on EEG changes in sleeping stages 2 to 4 and REM as well as a noticeable increase in total sleep time and reduction in the number of awakenings. Clinical studies have shown that this drug at dosages of 15–30 mg displays similar efficiency as flurazepam and triazolam with the significantly reduced rebound of insomnia after withdrawal, which often occurs in the case of triazolam (Ankier & Goa, 1988a; Kales, 1990). This advantage is probably associated with carryover hypnotic effects of quazepam after treatment termination because of the same or longer elimination half-lives of active metabolites than the parent drug. Nevertheless, long-term use of quazepam may result in tolerance or dependence, and effects on psychomotor skills (Ankier & Goa, 1988a). Therefore, it can be said that quazepam is the drug of choice for the short- or medium-term treatment of insomnia when withdrawal events are particularly irritating.

Pharmacodynamics: Detailed quazepam mechanism of action has not been fully understood. Selective binding of benzodiazepines such as quazepam with a trifluoroethyl group on position 1 to the benzodiazepine receptor type I (BZ₁) is a common phenomenon, where the lack of ataxia with quazepam observed in several animal studies may be originated from this behavior (Iorio et al., 1984). There are several general hypotheses about the pharmacodynamic behavior of quazepam in the cerebral cortex, substantia nigra, and the cerebellum. Since BZ₁ receptors are attached to

the GABA receptor complex, the binding of quazepam to these receptors could induce GABA transmission (Corda et al., 1986). Furthermore, quazepam seems to reduce cyclic 3',5'-guanosine monophosphate (cGMP) concentrations in the neurons via promoting GABA transmission, leading to pentylenetetrazol-induced convulsions in the brain (Biggio et al., 1977; Ongini et al., 1982). Some animal studies have also described that quazepam may activate sleep-inducing and synchronizing hypnogenic mechanisms at the level of bulbo-pontine reticular neurons in the brain stem (Mariotti & Ongini, 1983; Ongini et al., 1982).

Metabolism: Quazepam is extensively metabolized by the CYP3A4 enzyme in the liver, but at a slower rate than other benzodiazepine hypnotics such as flurazepam and triazolam. The first metabolic derivative of quazepam is 2-oxoquazepam produced from the substitution of sulfur by oxygen. This pharmacologically active metabolite then undergoes either *N*-dealkylation or hydroxylation to form *N*-desalkyl-2-oxoquazepam or 3-hydroxy-2-oxoquazepam, respectively (Fig. 5.11). Conjugation of these metabolites with glucuronic acid finally occurs followed by their excretion in the urine (Zampaglione et al., 1985). Quazepam and 2-oxoquazepam are the main species in plasma that reach their maximum concentrations within the first 2 h after taking the medication. In contrast, *N*-desalkyl-2-oxoquazepam appears in plasma more slowly, reaching its peak concentration within 14 h, and stays longer in the body. Therefore, quazepam and 2-oxoquazepam could be responsible for sleep induction, and *N*-desalkyl-2-oxoquazepam with a longer half-life (Table 5.22) may contribute to the lack of rebound insomnia (Zampaglione et al., 1985).

Pharmacokinetics and dosage: Quazepam is commercially available for oral administration as 15 mg Doral tablets. Following a single 15 mg oral dose of quazepam, a peak plasma concentration of approximately 30 µg/L is reached within 2.5 h (Ankier & Goa, 1988b). It is extensively bound to plasma protein (> 95%) with a rapid onset of action of 30 min. Interestingly, the plasma concentration of the drug is significantly lower when taken in the morning (21 µg/L) compared with bedtime (32.3 µg/L), which is probably due to changes in hemodynamics related to sleep or the supine position at night time (Hilbert et al., 1984). Following the 25 mg oral dose of the medication, the 2-oxoquazepam active metabolite appears in plasma with the maximum plasma concentration of 45 µg/L within 1.6 h, while *N*-desalkyl-2-oxoquazepam is formed more slowly reaching 50 µg/L within 14 h (Table 5.22). The elimination half-life of quazepam varies from 25 to 47 h following a single 15 mg oral dosage in young individuals (27 years

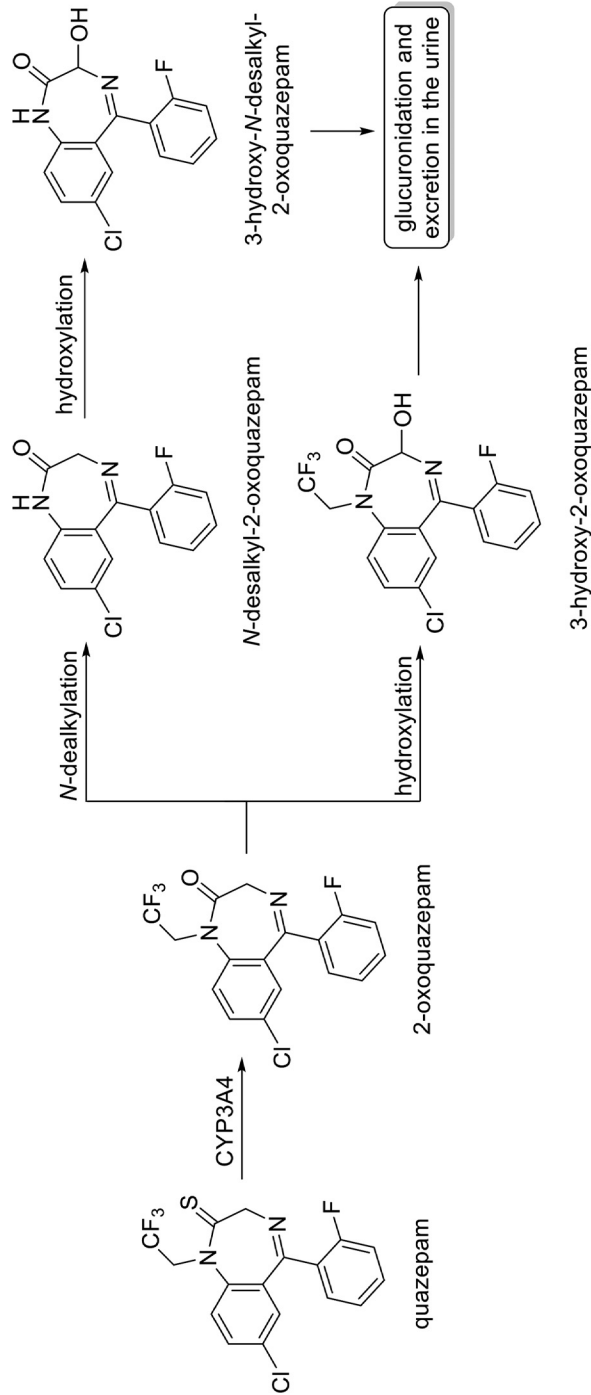


Figure 5.11 Metabolic pathways of quazepam.

Table 5.22 Pharmacokinetic parameters of quazepam and its active metabolites after single 15 mg dose.

Protein albumin binding	> 95%
Onset of action (min)	30
$C_{\max}$ (μg/L)	29–31
$T_{\max}$ (h)	2.5–2.7
V_d (L/kg)	5–8.6
AUC_{0-28} (μg/L × h)	509
AUC_{0-120} (μg/L × h)	275–320
$t_{1/2}$ (h)	25–53.3
Quazepam ^a	$C_{\max} = 148 \text{ μg/L}$, $T_{\max} = 1.5 \text{ h}$, $t_{1/2} = 39 \text{ h}$
2-Oxoquazepam metabolite ^a	$C_{\max} = 45 \text{ μg/L}$, $T_{\max} = 1.6 \text{ h}$, $t_{1/2} = 40 \text{ h}$
N-Desalkyl-2-oxoquazepam metabolite ^a	$C_{\max} = 50 \text{ μg/L}$, $T_{\max} = 14 \text{ h}$, $t_{1/2} = 69 \text{ h}$

^a After 25 mg oral dose.

old). In contrast, this value is slightly longer (53.3 h) in the elderly patients that results from their impairment in hepatic oxidative enzymes such as CYP3A4 and less capability in bio-transforming the drug. Quazepam is widely distributed in the body tissues with a large volume of distribution (5–8.6 L/kg), reflecting the high lipophilicity of the drug ($\log P = 403$). The recommended initial dose of quazepam is 15 mg prior to bedtime with the possibility to increase to 30 mg, which provides hypnotic effects in adults with chronic or transient insomnia ([The US Food and Drug Administration, 2016a](#)). Although the onset of hypnotic effects is often observable after the initial dosage, the 30 mg dose may induce sleep much quicker than the 15 mg dose. In elderly people, the initial quazepam dose of 15 mg is advised to be reduced to 7.5 mg, as this patient group may be more sensitive to the medication ([The US Food and Drug Administration, 2016a](#)).

Adverse effects: The most routine adverse effects of quazepam include drowsiness, headache, feeling very tired, dizziness, dry mouth, and upset stomach. This drug is generally well tolerated and side effects such as amnesia and confusion, which are the usual adverse reactions with triazolam and alprazolam, are very rarely with 30 mg quazepam administration and even have not been reported with 15 mg dosage ([Ankier & Goa, 1988b](#)). Kales and co-workers have shown that sleepiness induced by 30 mg quazepam is more frequent and intense than 30 mg flurazepam ([Kales et al., 1986a](#)). The common adverse events of memory impairment, hyperexcitability, and psychotic syndromes that occurred with the administration of triazolam

and alprazolam, have not been observed with 15 mg dose of quazepam or have been very rare with the 30 mg dosage (Kales et al., 1986a). As mentioned earlier, quazepam causes no withdrawal syndromes of rebound insomnia, which is so common in other benzodiazepines. Quazepam is widely distributed to the placenta and is present in breast milk (Kales et al., 1986a). According to the US FDA, quazepam is classified as Pregnancy Category C medication, which indicates that there are no adequate and well-controlled studies in pregnant women, even though the drug's adverse effects on the animal fetus are documented (The US Food and Drug Administration, 2016a). Administration of quazepam in animal studies prior to or during pregnancy has been shown to result in hypotonia, hypothermia, respiratory depression, and minor skeletal variations in infants. Therefore, it is highly advised that quazepam can be used during pregnancy only if the potential benefits justify the potential risks (Kales et al., 1986a).

Interactions: There is a lack of information regarding the potential quazepam interactions with alcohol or other medications. Nevertheless, according to the US FDA, quazepam may cause enhanced CNS depressant effects when co-administered with alcohol or other CNS depressants such as psychotropic medications, anticonvulsants, and antihistamines (The US Food and Drug Administration, 2016a).

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